

Virtuoso Layout Pro: T5 Interactive Routing

Course Version IC25.1

Lab Manual

Revision 1.0

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Printed in the United States of America.

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Module 1: About This Course

There are no labs for this module.

Module 2: Basic Concepts and Settings in Wire Creation

Lab 2-1 Using Wire Editing Options with the Create Wire Command

Objective: To explore the Wire Editing options with the Create Wire command.

In this lab, you

- ◆ Explore the most commonly used Create Wire options: Auto Merge Wires, Allow Loops, Adjust Edited Vias Parameters, and Show Alignment Markers.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. Untar the lab database file using the following command.

```
tar -xzvf <filename.tar.gz>
```

2. Update the following line in the `/VLPT5_IC_25_1/.cshrc_VLPT5` file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
```

3. Save and close the file.

4. Source the updated class setup file.

```
source .cshrc_VLPT5
```

Opening the Software

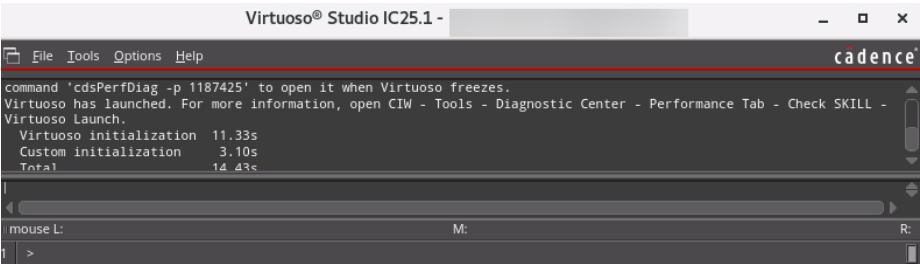
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./VLPT5_IC_25_1/M02_VLPT5_Bsc_concept
```

2. To start the Cadence® Virtuoso® Studio software, enter this command in the same terminal window:

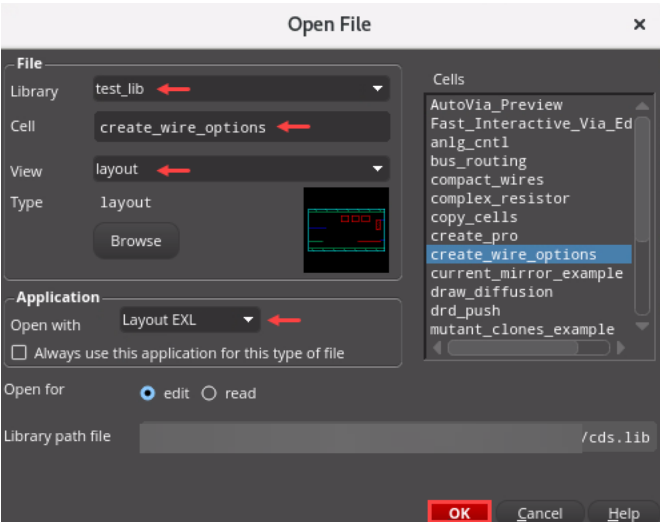
```
virtuoso &
```

The Command Interpreter Window (CIW) appears.



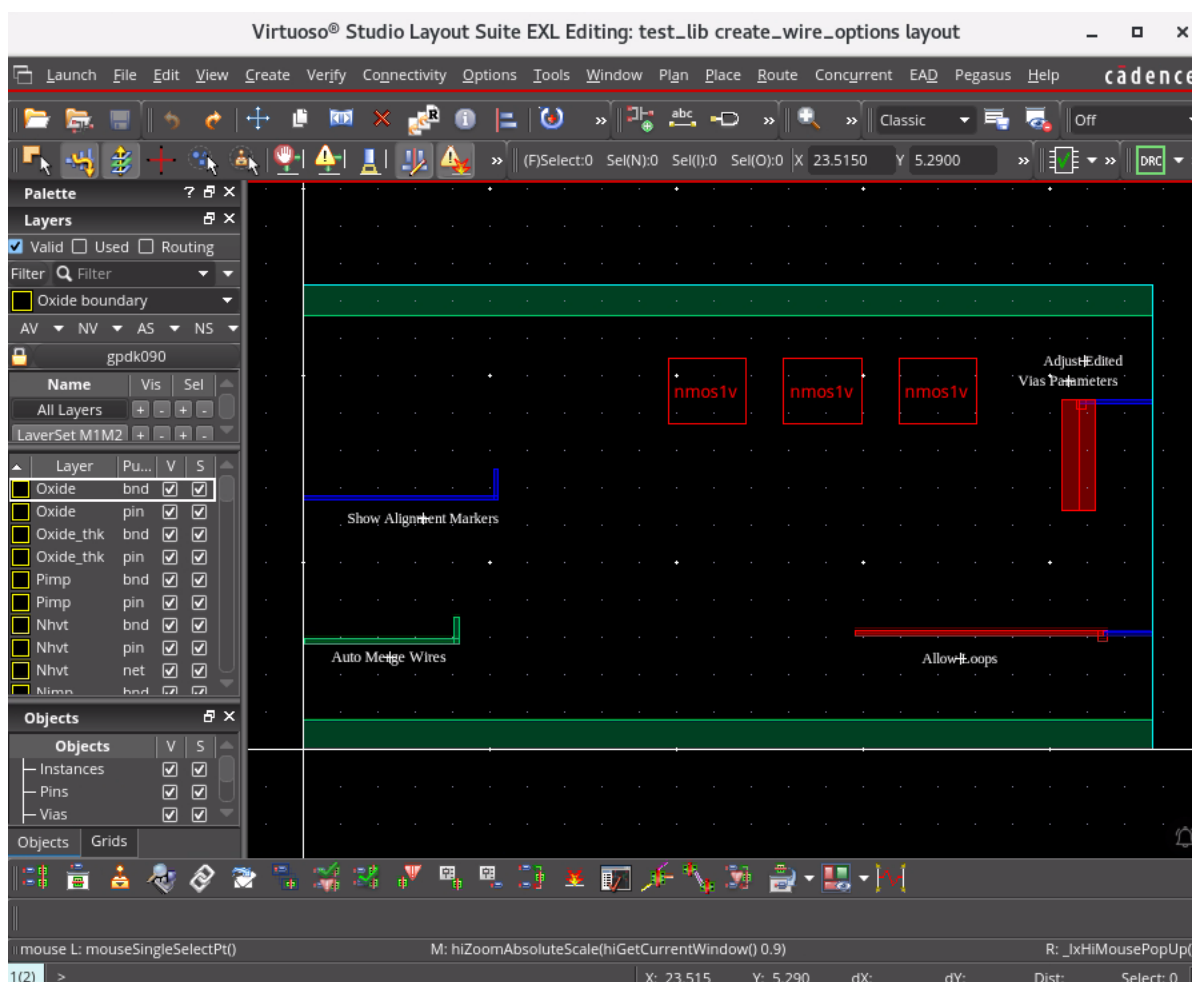
3. In the CIW, choose **File – Open** menu command and enter these values in the form.

Library	test_lib
Cell	create_wire_options
View	layout



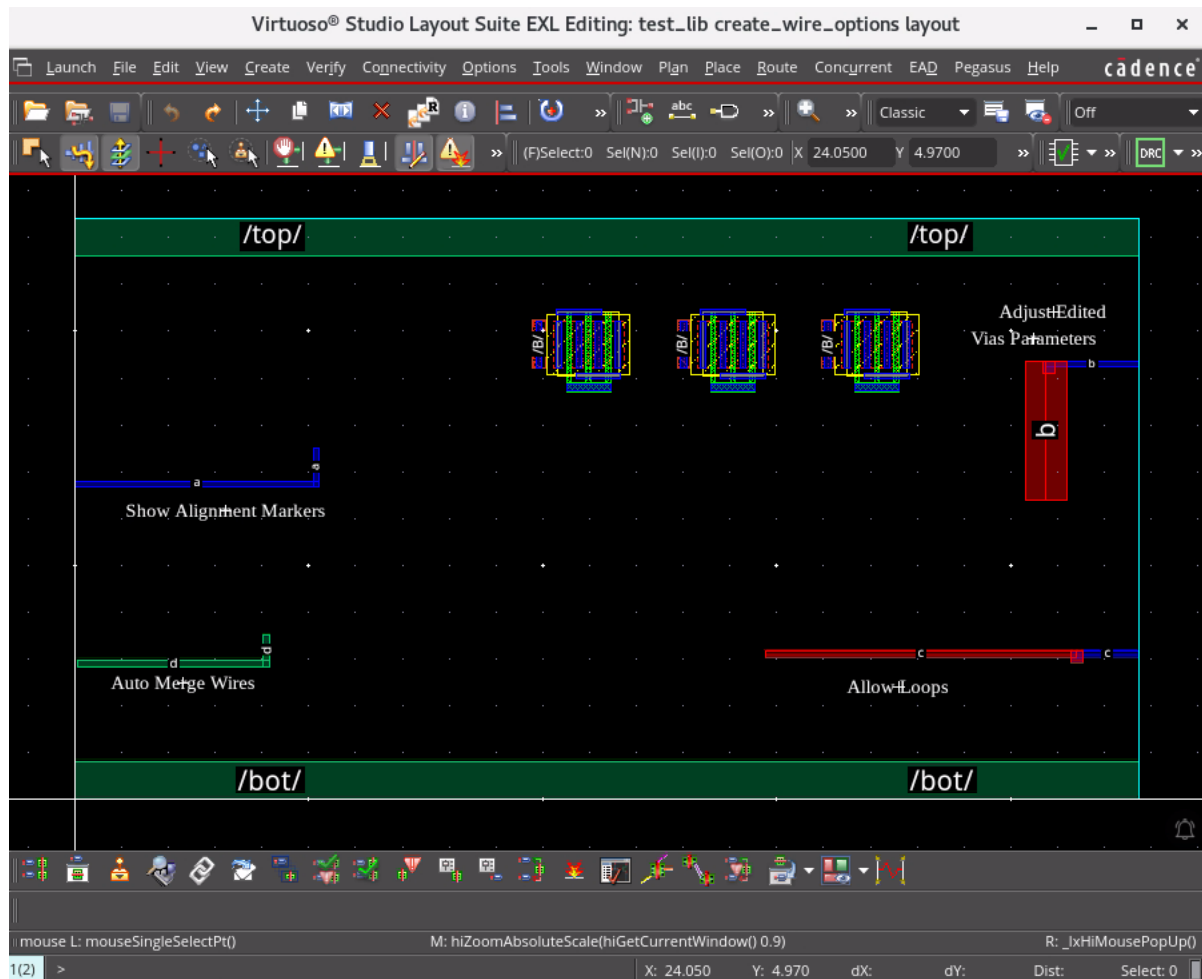
Both the schematic and layout views appear in EXL mode.

4. Minimize the schematic window and maximize the layout window.



5. In the layout window, do the following steps.
 - a. Press the **Shift+F** bindkey to view all the levels of the hierarchy in the canvas.
 - b. Disable the **Palette** assistant from the **Window – Assistants** menu to enlarge the canvas port.

- c. Choose the **Options – Net Name Display** menu command to invoke the Net Name Display Options form and uncheck the **Enable** button to disable the net name display.

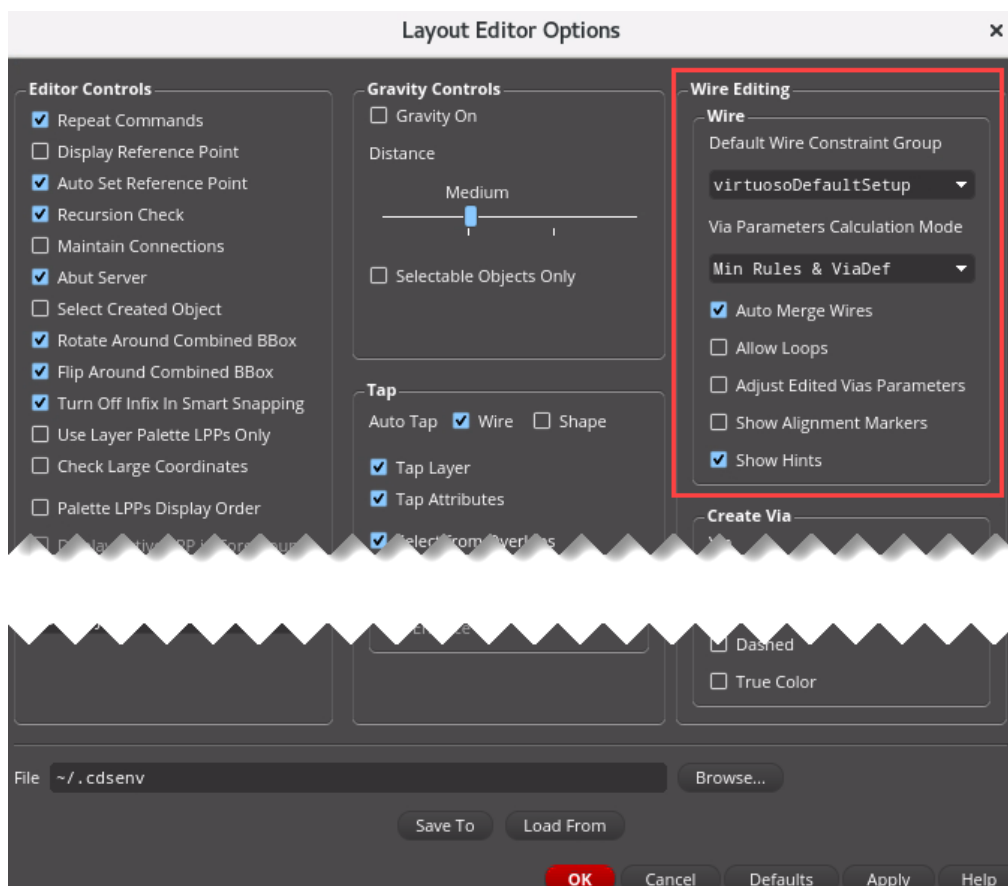


In the next steps, you explore the following options in the **Layout Editor Options** form:

- Auto Merge Wires
- Allow Loops
- Adjust Edited Vias Parameters
- Show Alignment Markers

6. In the layout canvas, choose the **Options – Editor** menu command or press the bindkey **Shift+E** to invoke the Layout Editor Options form.

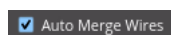
You will use the Wire Editing section in the Layout Editor Options form.



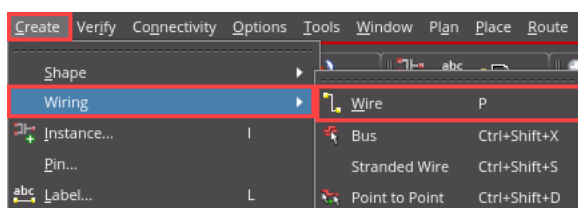
Auto Merging Wires

In the Layout Editor Options form, with the **Auto Merge Wires** option enabled, it automatically merges the connected paths and collinear pathSegs while creating or stretching a wire. The checkbox is selected by default. To prevent automatic merging of collinear (symbolic or geometric) wires, disable the **Auto Merge Wires** option.

1. With the **Auto Merge Wires** option **on** (default), do the following steps.



- a. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon from the Create toolbar or use the bindkey **P** and click on the **Metal3** pathSeg in the Auto Merge Wires section in the canvas.



- b. Create a wire from the Metal3 pathSeg, and you can observe that the wire automatically merges because the Auto Merge Wires option is on in the Layout Editor Options form.



- c. Press the Esc key to Cancel the Create Wire command.
 - d. Choose the Edit – Undo menu command or click the Undo icon in the Edit toolbar or press the bindkey U to undo the created wire.
2. With the **Auto Merge Wires** option **off**, do the following steps.



- a. Repeat the previous Step 1-a to invoke the **Create Wire** command.
- b. Repeat the previous Step 1-b and you will notice that the wire does not auto-merge because the **Auto Merge Wires** option is **off** in the Layout Editor Options form.



- c. Press the **Esc** key to **Cancel** the Create Wire command.
- d. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

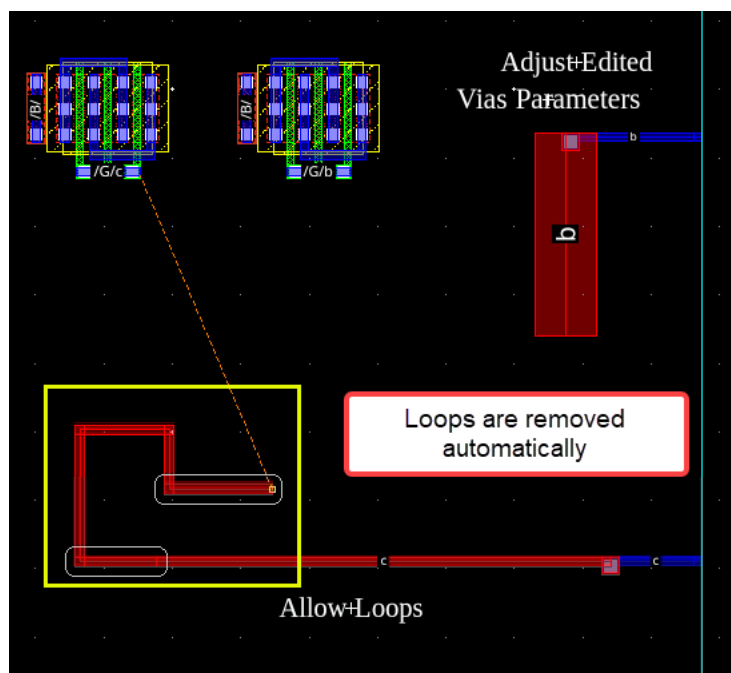
Allowing Loops

The “loops” can be automatically removed by the **Create Wire** command with the **Allow Loops** option disabled in the Layout Editor Options form. The checkbox is deselected by default. Although this can be useful in the top-level routing, this feature can be annoying at the device level, where there might be a good reason to create such loops, which can be obtained by enabling the **Allow Loops** option.

1. With the **Allow Loops** option **off** (default), do the following steps.



- a. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** and click on the **Metal2** pathSeg in the Allow Loops section in the canvas.
- b. Create a wire that loops, and you can observe that the “loops” are removed automatically while creating the wire because the **Allow Loops** option is **off** in the Layout Editor Options form.



- c. Press the **Esc** key to **Cancel** the Create Wire command.
- d. Choose the **Edit – Undo** menu command or click the **Undo** icon in the Edit toolbar or press the bindkey **U** to undo the created wire.

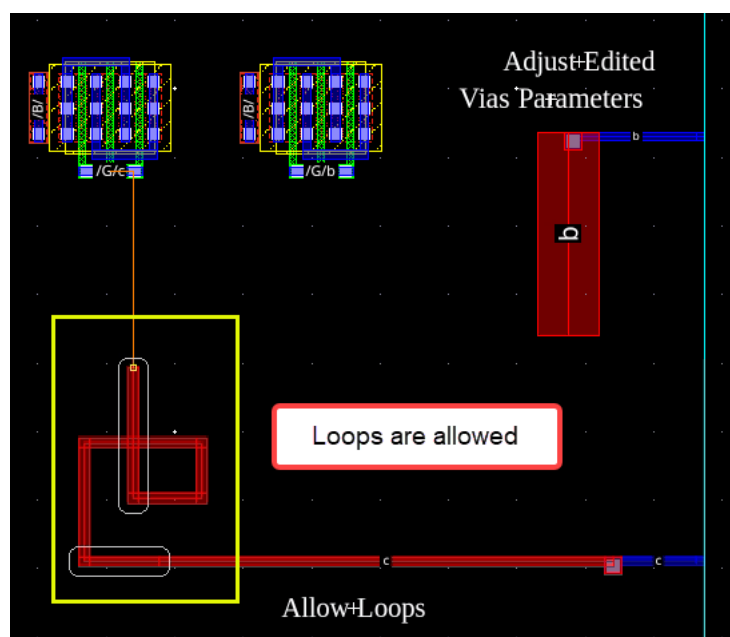
2. With the **Allow Loops** option **on**, do the following steps.



- a. Repeat the previous Step 1-a to invoke the Create Wire command.

- b. Repeat the previous Step 1-b and recreate the Metal2 wiring.

This time, the “loops” stay unchanged because the **Allow Loops** option is **on** in the Layout Editor Options form.



- c. Press the **Esc** key to **Cancel** the Create Wire command.
- d. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

Adjusting Edited Vias Parameters

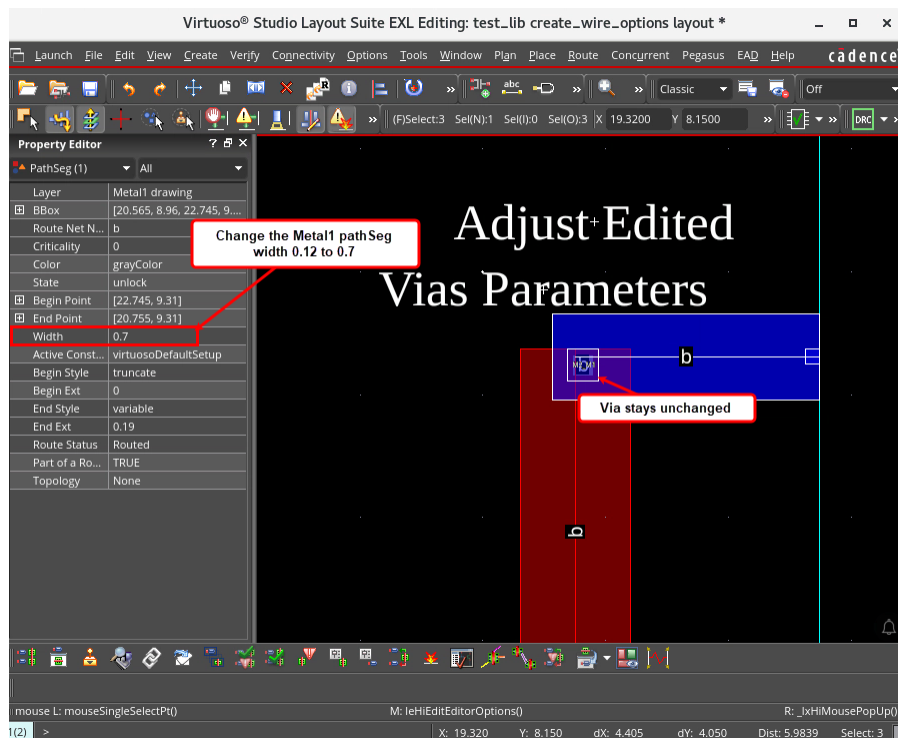
In the Layout Editor Options form, with the **Adjust Edited Vias Parameters** option disabled, when the size of a wire element (pathSeg) is changed, the via stays unchanged and the via calculation is not triggered. The checkbox is deselected by default. To retrigger the via calculation when the size of a wire element (pathSeg) is changed, enable the **Adjust Edited Vias Parameters** option.

1. With the **Adjust Edited Vias Parameters** option **off** (default), do the following steps.



- a. In the layout canvas, choose the **Window – Assistants – Property Editor** menu command to display the Property Editor assistant.
- b. In the design canvas, select the **Metal1** pathSeg in the Adjust Edited Vias Parameters section.

- c. In the Property Editor assistant, change the **Metal1** pathSeg width from **0.12** to **0.7**.



The via remains unchanged (single cut) even if there is enough place to use an array of vias because the **Adjust Edited Vias Parameters** option is **off** in the Layout Editor Options form.

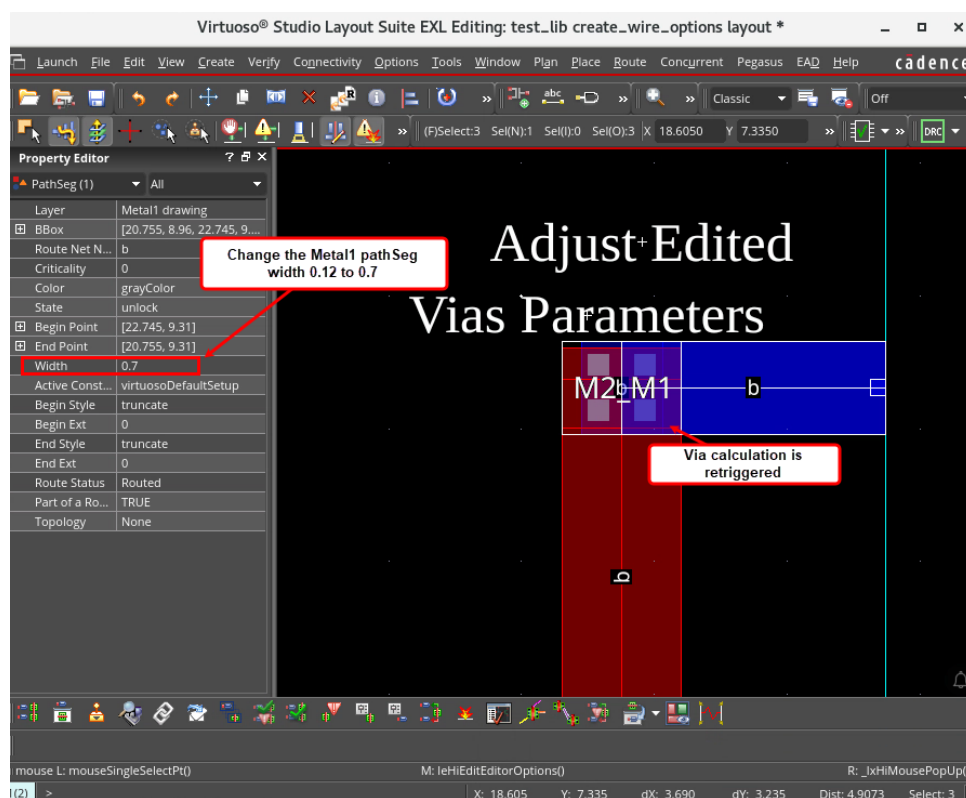
- d. Undo the previous edits.

2. With the **Adjust Edited Vias Parameters** option **on**, do the following steps.



- a. Repeat the previous Step 1-b to select the Metal1 pathSeg in the Adjust Edited Vias Parameters section.
- b. Repeat the previous Step 1-c and change the Metal1 pathSeg width again from 0.12 to 0.7.

This time the via calculation is triggered and a larger via (an array of 2x2 cuts) is used because the **Adjust Edited Vias Parameters** option is **on** in the Layout Editor Options form.



- c. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

Showing Alignment Markers

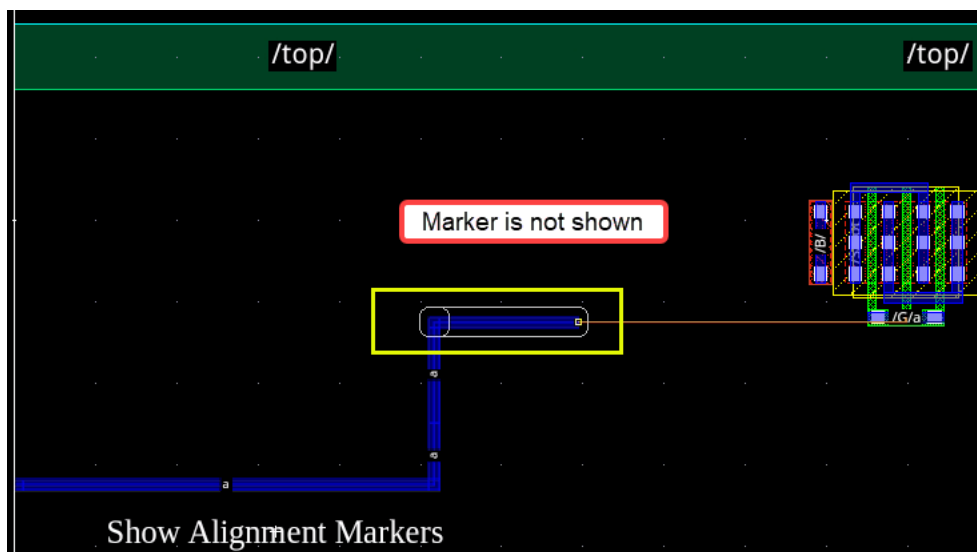
The alignment markers can be very useful when routing to the target pins, which are “out of sight”. In this case, when the wire is aligned to the target pin, a small white arrow is shown. In the Layout Editor Options form, with the **Show Alignment Markers** option disabled, when the wire is aligned to the target pin, the marker is not visible. The checkbox is deselected by default. To view the marker when creating the wire, enable the **Show Alignment Markers** option.

1. With the **Show Alignment Markers** option **off** (default), do the following steps.

☐ Show Alignment Markers

- a. In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar, or use the bindkey **P** and click on the **Metal1** pathSeg in the Show Alignment Markers section in the canvas.
- b. Create a wire from the **Metal1** pathSeg and move the mouse pointer upwards.

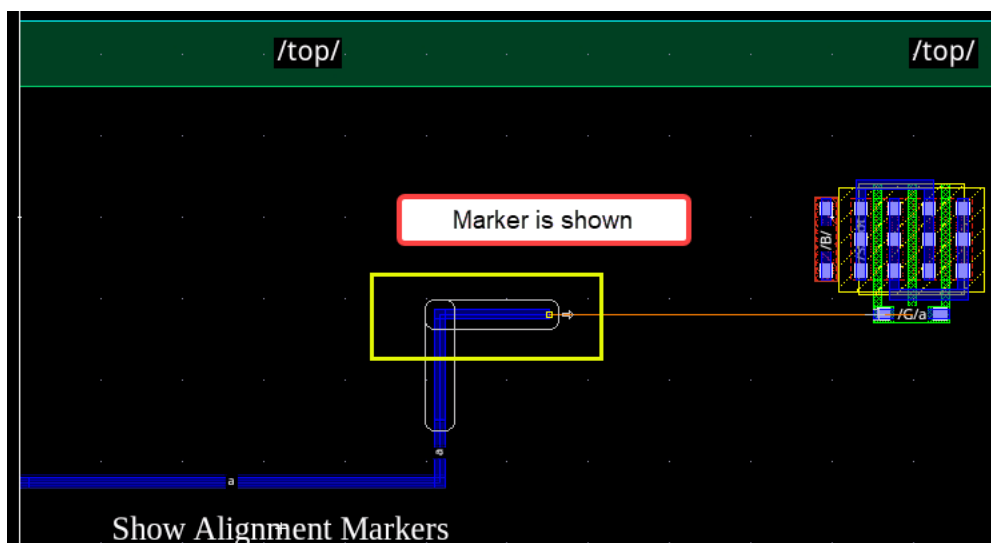
No marker is visible because the **Show Alignment Markers** option is **off** in the Layout Editor Options form.



- c. Press the **Esc** key to cancel the **Create Wire** command.
 - d. Choose the **Edit – Undo** menu command or click the **Undo** icon in the Edit toolbar or press the bindkey **U** to undo the created wire.
2. With the **Show Alignment Markers** option **on**, do the following steps.



- a. Repeat the previous **Step 1-a** to invoke the **Create Wire** command.
- b. Repeat the previous **Step 1-b**. This time when you move the mouse pointer up, as soon as the wire is aligned with the target pin, a small white marker is shown because the **Show Alignment Markers** option is **on** in the Layout Editor Options form.



- c. Press the **Esc** key to **Cancel** the Create Wire command.
 - d. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.
3. Close the schematic and layout views. Do **NOT** save.
 4. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you

- ◆ Used the Auto Merge Wires option
- ◆ Used the Allow Loops option
- ◆ Used the Adjust Edited Vias Parameters option
- ◆ Used the Show Alignment Markers option



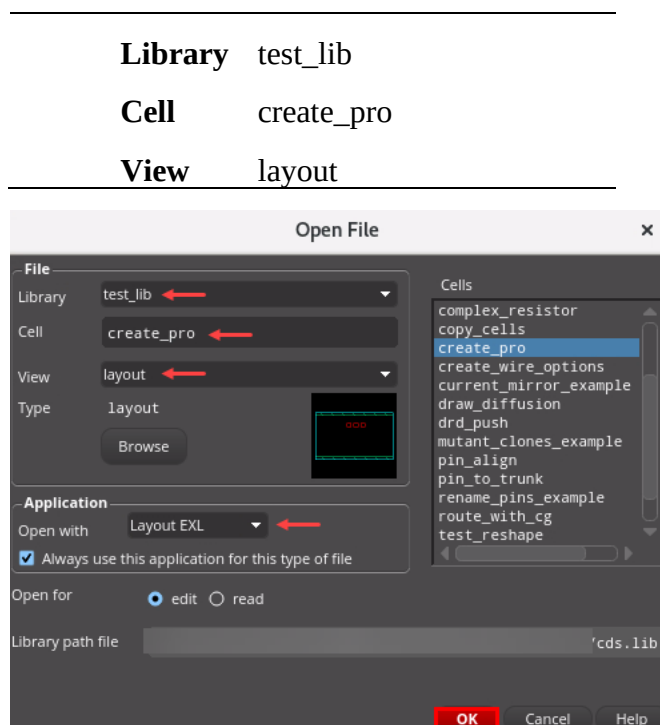
Lab 2-2 Using Override Constraints in the Wire Assistant

Objective: To use the existing and create new Process Rule Overrides in the Wire Assistant.

In this lab, you

- ◆ Explore how to populate the **Override Constraints** section with existing **Process Rule Overrides (PRO)** from the **Seed Attributes From a Constraint Group** option.
- ◆ Create new **Process Rule Overrides (PRO)** using the **Wire Assistant**.
- ◆ Experiment switching between different constraint groups.

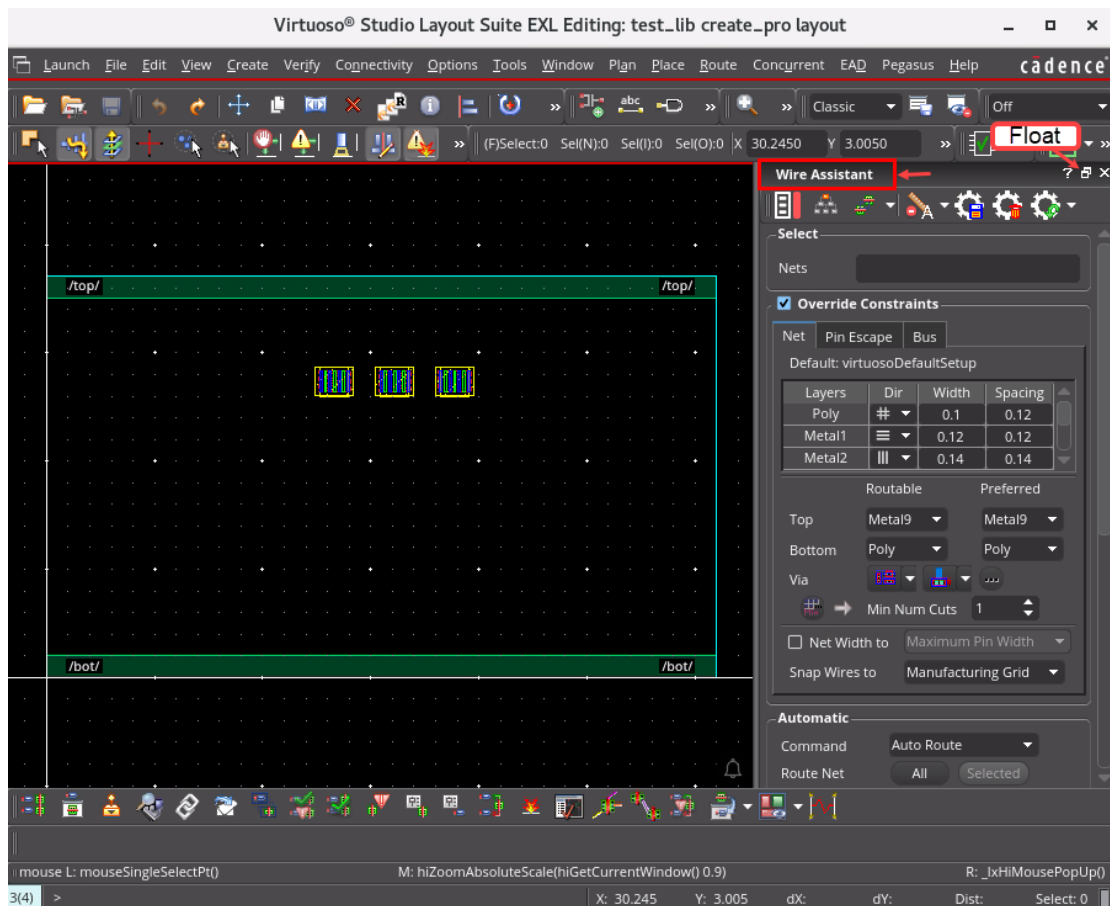
1. In the CIW, choose **File – Open** menu command and enter these values in the form.



Both the schematic and layout views appear in EXL mode.

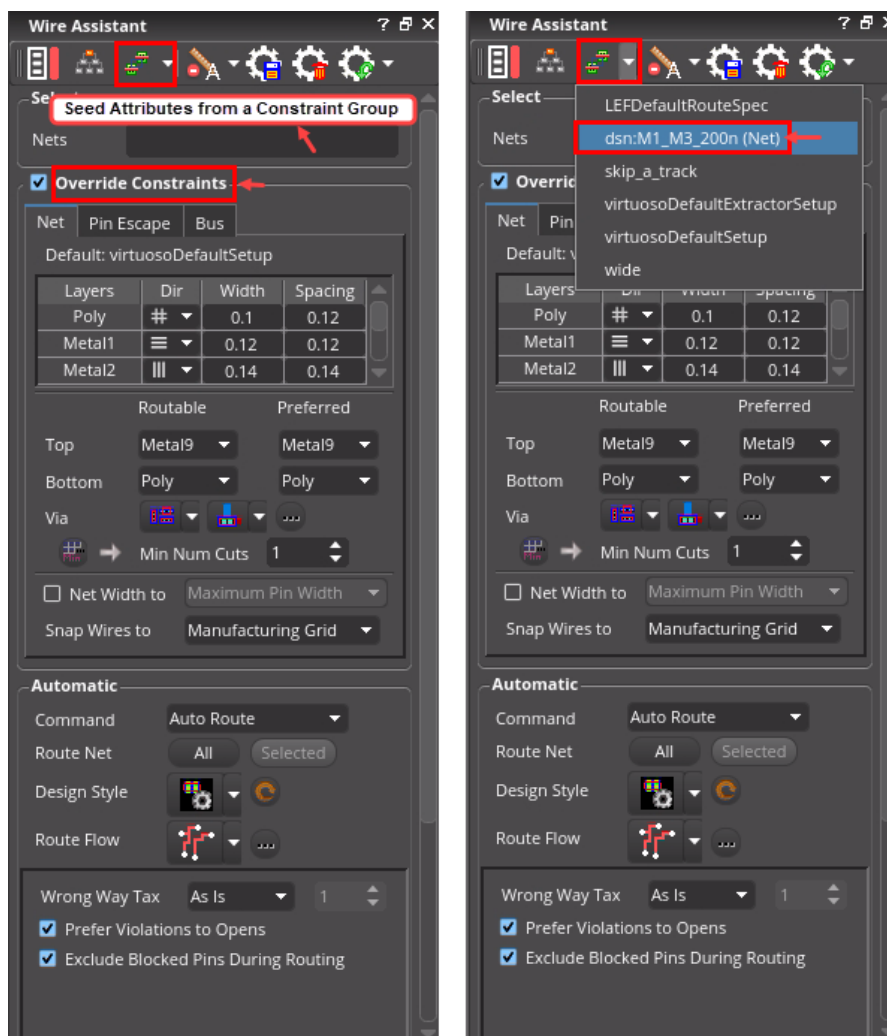
2. Minimize the schematic window and maximize the layout window.
3. In the design window, do the following steps:
 - a. Press the **Shift+F** bindkey to view all the levels of the hierarchy in the canvas.
 - b. Disable the **Palette** assistant from the **Window – Assistants** menu to enlarge the canvas port.

4. In the layout canvas, choose the **Window – Assistants – Wire Assistant** menu command to display the Wire Assistant.
 - a. Click the **Float** button in the **Wire Assistant** so that it can be undocked.

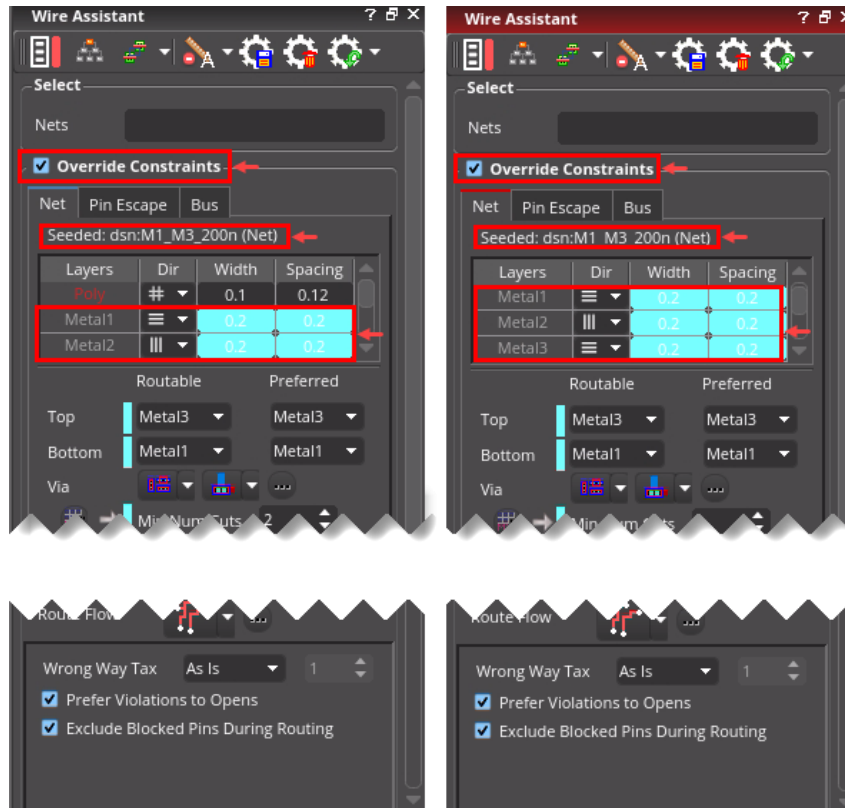


Using Existing PRO to Populate the Override Constraints Section in the Wire Assistant

1. From the **Seed Attributes From a Constraint Group** drop-down list on the **Wire Assistant** toolbar, select the **dsn:M1_M3_200n (Net)** constraint group to populate the **Override Constraints** section from the **virtuosoDefaultSetup** constraint group with the new overridden values (**PRO**), as shown here.



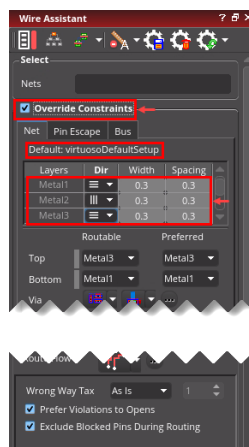
The **Override Constraints** section displays the new set of rules (**PRO**) of the **dsn:M1_M3_200n (Net)** constraint group, as shown here.



Creating New PRO Dynamically to Populate the Override Constraints Section in the Wire Assistant

1. Modify the rules in the **Override Constraints** section, as in this example.

Width and **Spacing** values of **Metal1**, **Metal2**, and **Metal3** to **0.3**



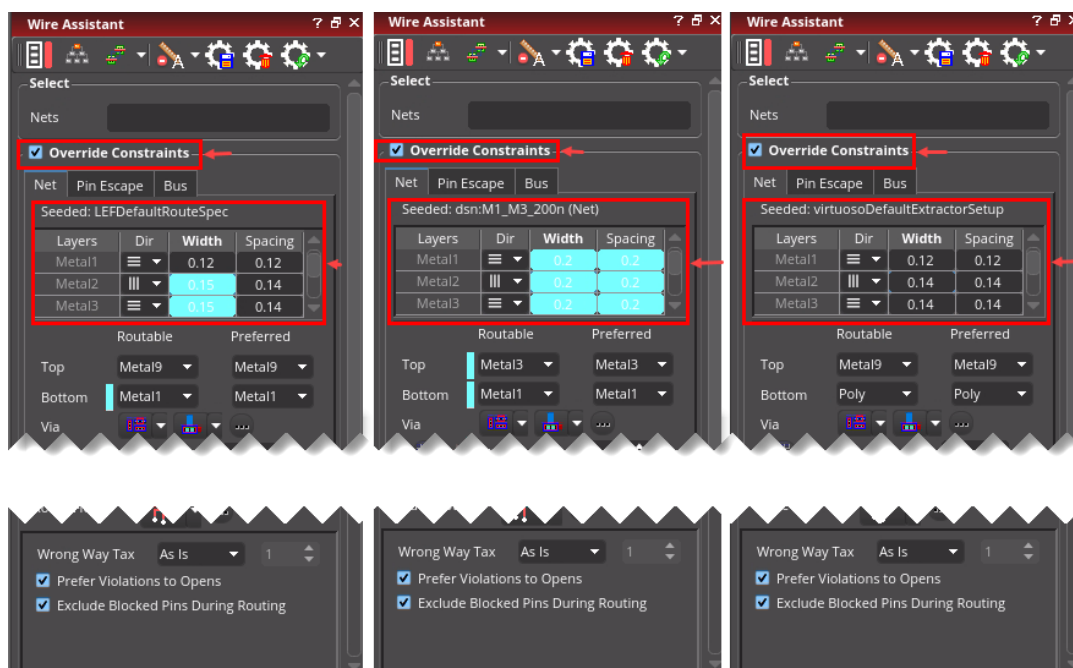
Note: You will see how to route with the dynamically created PRO in the later labs.

Switching Between Different Constraint Groups

1. Experiment switching between these three different sets of constraint groups.

- LEFDefaultRouteSpec
- dsn:M1_M3_200n (Net)
- virtuosoDefaultExtractorSetup

Every set of rules bring in different values for the **Width** and **Spacing** values of the layers in the **Override Constraints** section, as shown here.



2. Close all the cellviews without saving any changes you have made.



The prefix in the Process Rule Overrides constraint group names can be used to identify where they are stored: The letters “dsn” identify PRO which are stored in the layout design cellview.

3. Close any open forms and cellviews.
4. Leave the Virtuoso session.
5. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you

- ◆ Populated the Override Constraints section with existing Process Rule Overrides
- ◆ Created new Process Rule Overrides (PRO) using the Wire Assistant
- ◆ Experimented with switching between different constraint groups



Module 3: Creating and Modifying Wires

Lab 3-1 Connecting to Target Pins with the Create Wire Command

Objective: To use the **Create Wire** command with pin routes.

In this lab, you

- ◆ Explore how to use the **Create Wire** command to create a wire starting from a pin.
- ◆ Observe how the **Gravity Controls** setting in the Layout Editor Options form will affect the cycling through the possible targets.

Lab Setup

1. Follow the steps below to set up the environment and perform the lab:
2. If you continue from the previous lab, move to the sub-heading: Starting the session
Otherwise, proceed to the next step.
3. Update the following line in the `/VLPT5_IC_25_1/.cshrc_VLPT5` file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
```

4. Save and close the file.
5. Source the updated class setup file.

```
Source .cshrc_VLPT5
```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./VLPT5_IC_25_1/M03_VLPT5_Createandmodify
```

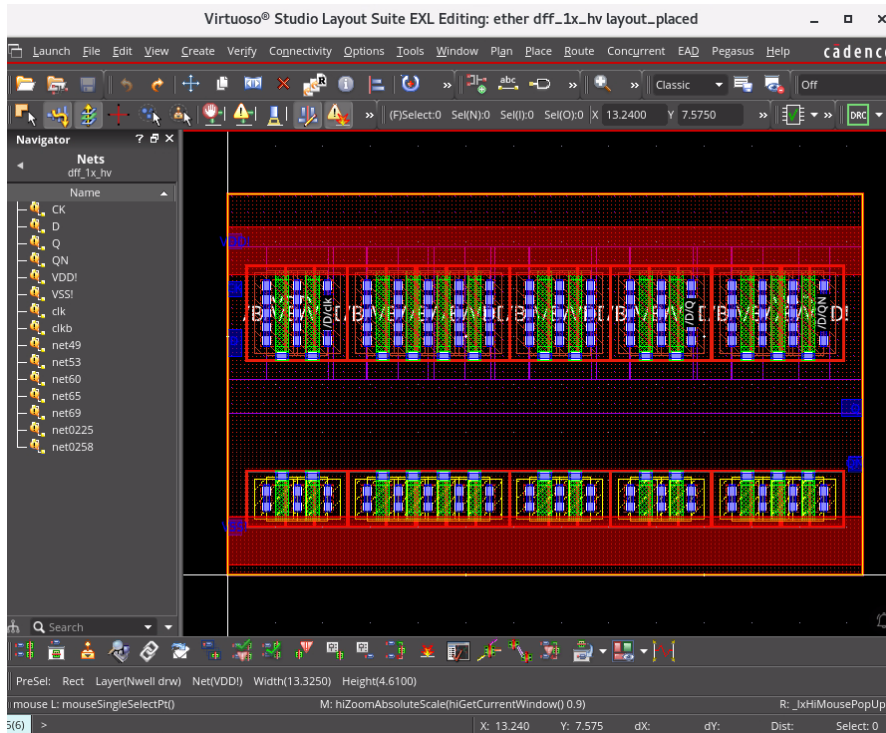
2. To start the Cadence® Virtuoso® Studio software, enter this command in the same terminal window:

```
virtuoso &
```

3. In the CIW, choose **File – Open** and enter the following values in the form.

Library	ether
Cell	dff_1x_hv
View	layout_placed

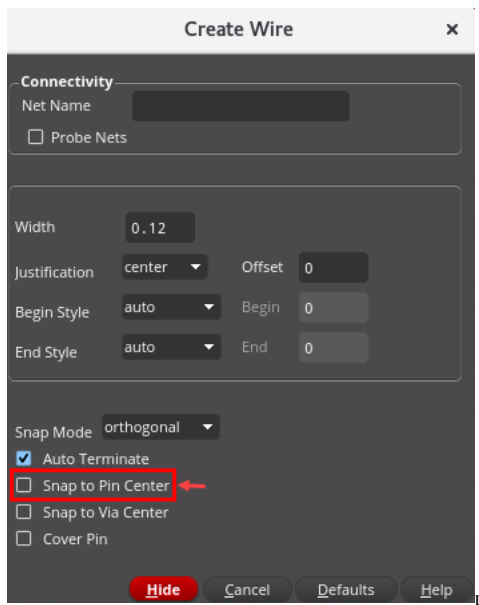
- In the canvas, choose **Launch – Layout EXL** to open the layout in EXL mode.
- In the design window, set the **OBJECTS** category in the **Navigator** to **Nets**.



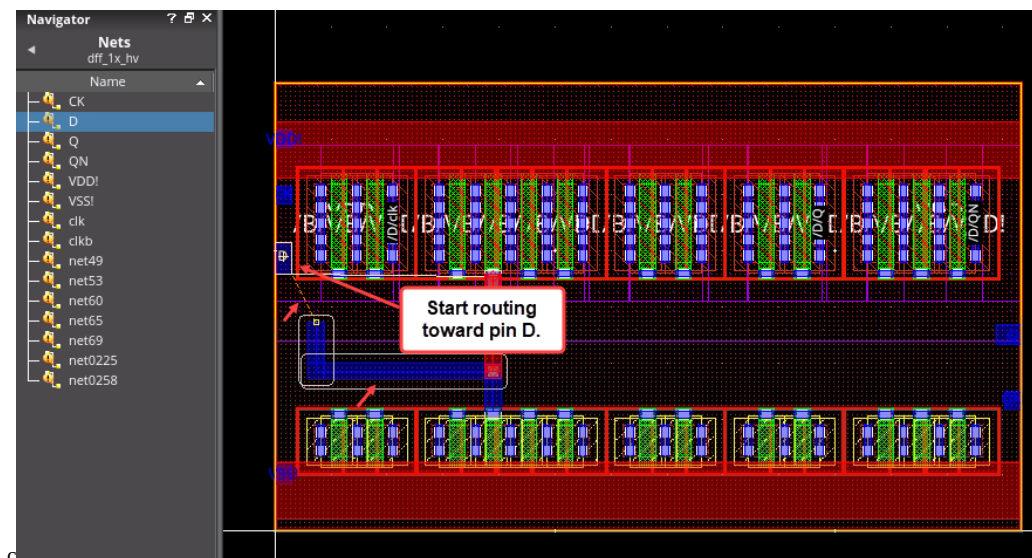
Note: Press the **Shift+F** bindkey to view all the levels of the hierarchy in the canvas.

- Select the net **D** in the **Navigator** assistant.
The pin and the flightlines highlight when you select the net **D** in the **Navigator**.
- In the design canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to invoke the Create Wire form.
If the Create Wire form does not pop up, press the **F3** key to bring up the form.

8. Make sure the **Snap to Pin Center** option is **off**, as shown here.

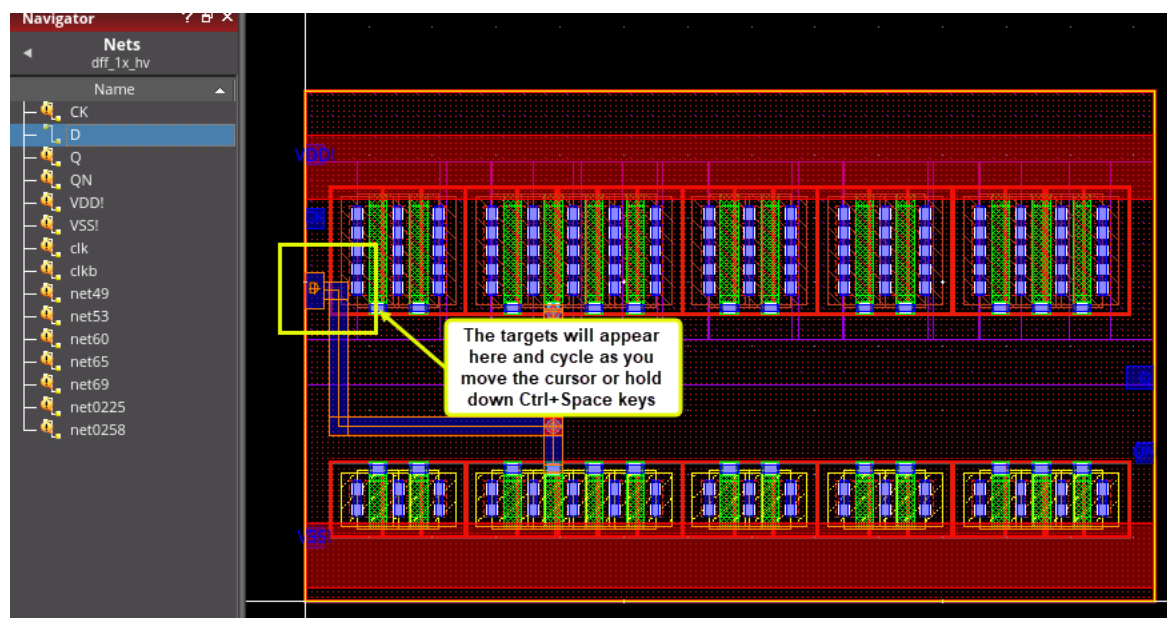


9. Start routing from either the **NM0** or the **PM0** device toward pin **D**, as shown here.



10. As you near pin **D**, you see the possible connecting targets appear. As you move your mouse pointer around the pin, the targets will cycle.

You can also use the **Ctrl+Space** key combination to cycle the targets.

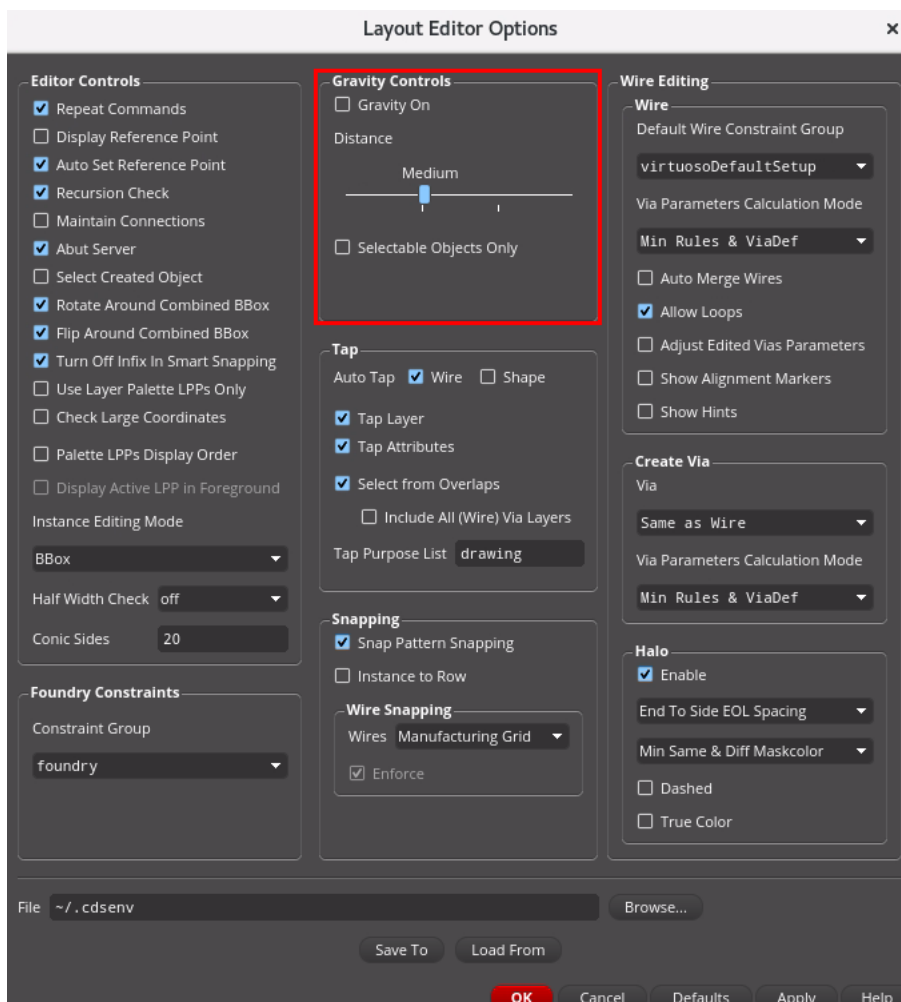


Note: **Gravity Controls: Gravity On** sets the drawing pointer to automatically snap to objects in the cellview. The pointer snaps to all the visible objects at any hierarchy level. Environment variable: *gravityOn*

Note: **Distance** slider sets the snap distance between the pointer and an object. You can choose from *Small*, *Medium*, *Large*, or *Very Large* using the slider. Environment variable: *gravityDistance*

11. Choose the **Options – Editor** menu command or press the bindkey **Shift+E** from the layout window to open the Layout Editor Options form.

12. Explore the **Gravity Controls** section with **Gravity On** selected/unselected and different **Distance** sliders set in the Layout Editor Option form when creating wires.



13. Discard the edits and do **NOT** close the layout view. You will use it in the next lab.

Summary

In this lab, you

- ♦ Used the Create Wire command to create a wire starting from a pin
- ♦ Used the Gravity Controls setting in the Layout Editor Options form to cycle through the possible targets



Lab 3-2 Using the Interactive Via Alignment Feature

Objective: To create wire using the interactive via alignment feature.

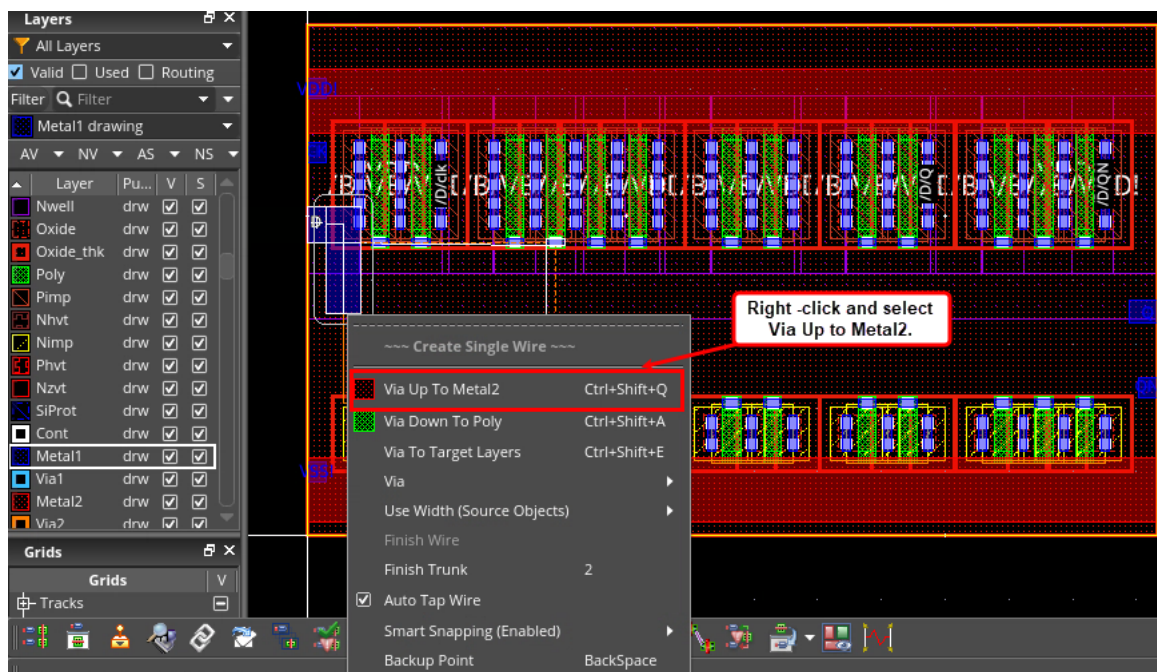
Starting with IC25.1, the wire editor can align the vias as they are created during the **Create Wire** or **Create Bus** commands. This interactive alignment provides options for automatic or custom alignment. In addition, assisted commands such as **Point to Point (P2P)**, **Finish Wire**, and **Finish Bus** now have an option for post-routing by alignment at the moment of command completion.

In this lab, you

- ◆ Explore the Select Via form.
- ◆ See how to change the orientation of the via using the middle mouse button.
- ◆ Use the **Ctrl + Mouse Wheel** to change the via offset point.

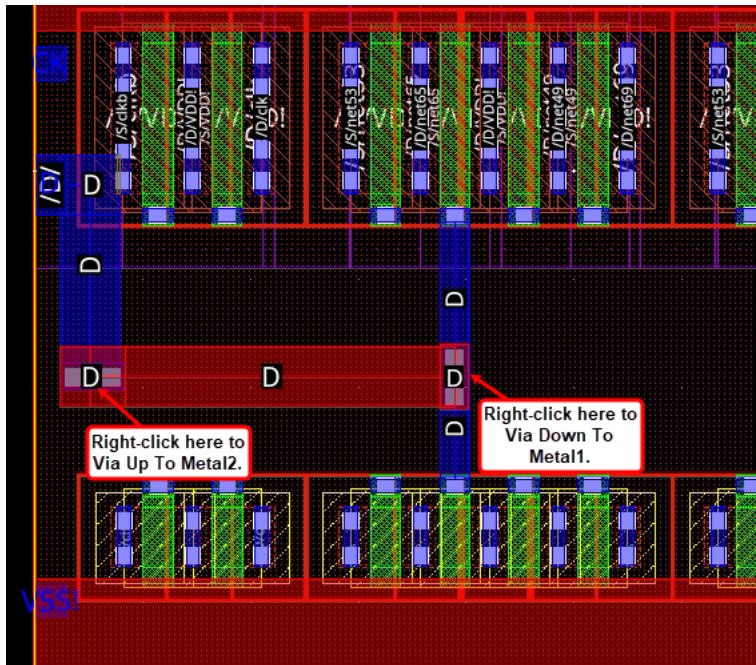
Creating Wire Using the Via Up, Via Down, and Finish Wire Options

1. Start with the open layout from the previous lab. If it is not open, then reopen it and create a wire from the pin **D** to one of the device connections.
2. When changing layers, **right-click** and select **Via Up** on the **Create Single Wire** context-sensitive menu. You can **middle-click** to rotate the via if needed.



3. To finish the connection, **right-click** and select **Via Down**. You can **middle-click** to rotate the via if required.
4. **Right-click** and select **Finish Wire** to end the wiring.

Notice how the vias created have the center of the layer enclosures matching the pathSeg centerline, coinciding endpoints.

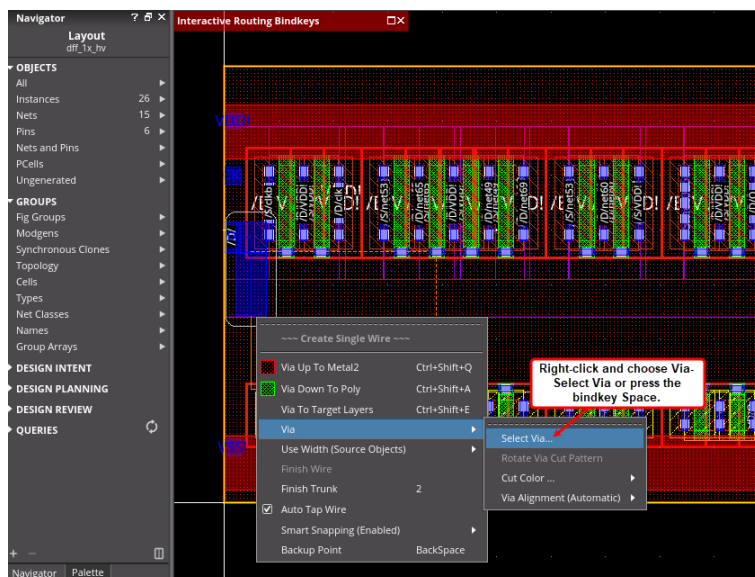


5. Delete this entire wire.

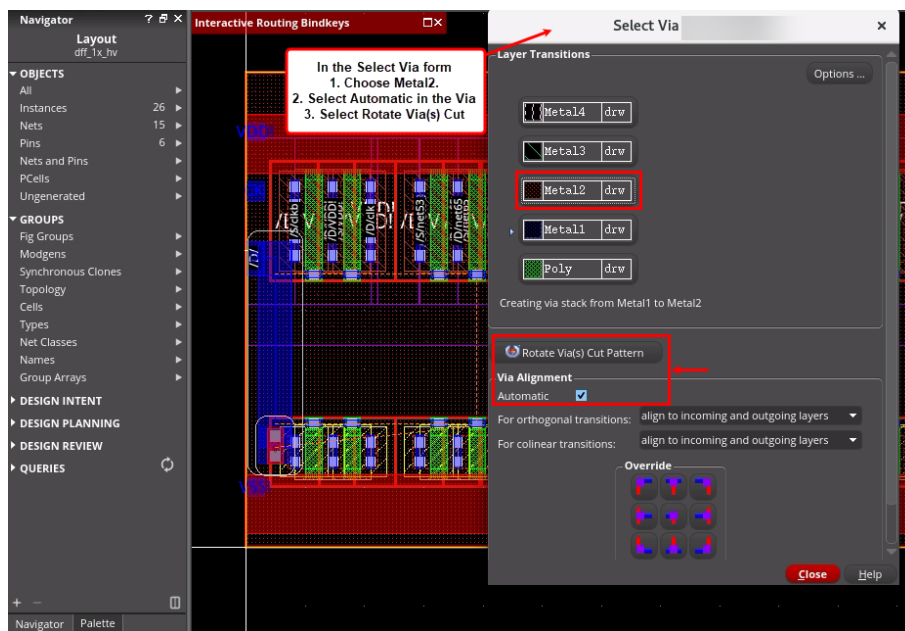
Creating Wire Using the Select Via Option

1. Create a new wire from the same pin **D**.

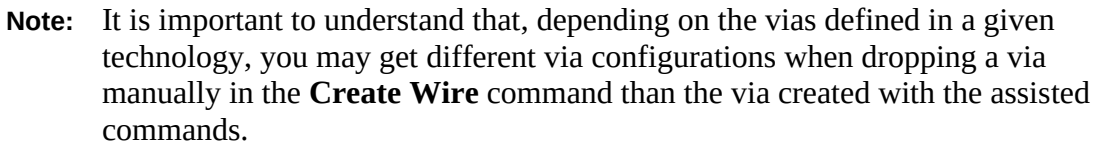
- When changing layers, **right-click** and choose the **Via – Select Via** menu entry or press the bindkey **Space** to invoke the Select Via form.



- Choose **Metal2** in the *Layer Transitions* section of the Select Via form.
- Select **Automatic** in the *Via Alignment* section of the Select Via form.
- Select **Rotate Via(s) Cut Pattern** if required.



- Alignment and rotation of the vias coincide with the direction of the routing.

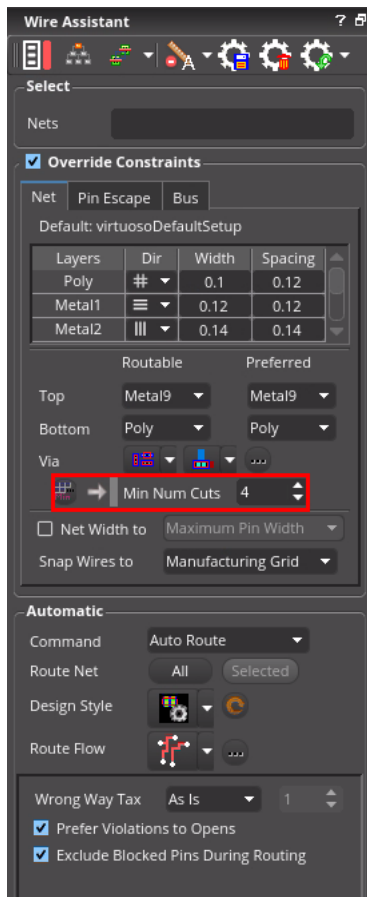


- ## Creating Wire by Specifying the Min Num Cuts Value in the Wire Assistant

- 34

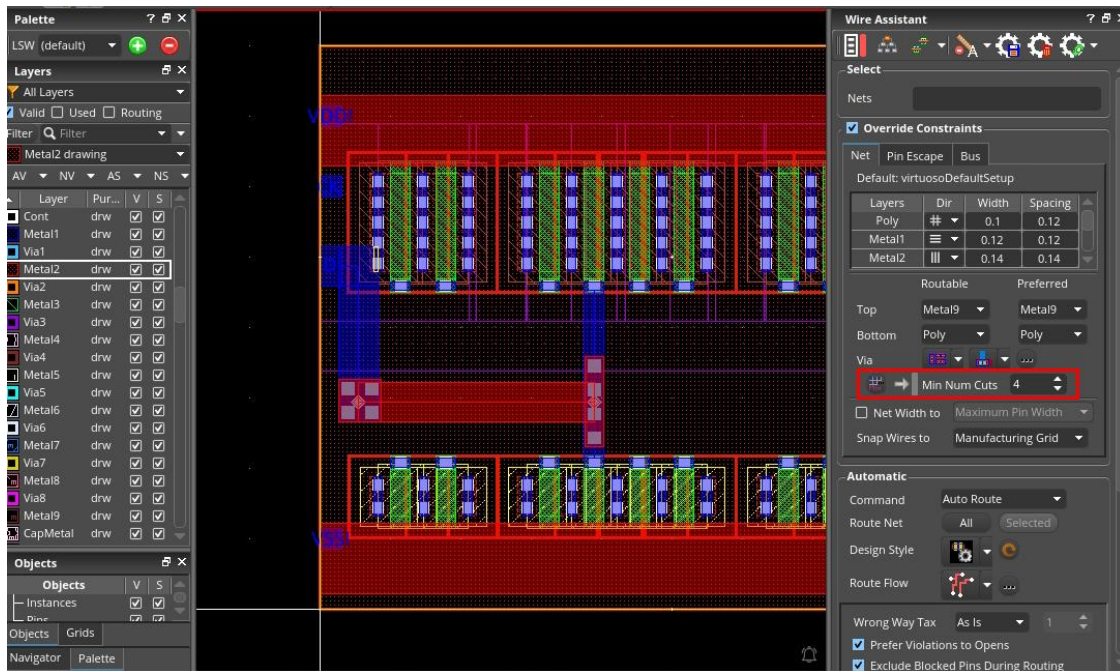
2. Set the value of **Min Num Cuts** to **4** in the **Wire Assistant**.

This setting will force the **Create Wire** command to use at least **4** cuts in all the vias that will be created.



3. Create a new wire from the same pin **D**.
4. Place a via using **right-click** and **Via Up**.
5. Use the middle mouse button to switch between horizontal and vertical orientations.
6. Use **Ctrl + Mouse Wheel** to change the anchor point of the placed via.

7. Complete the connection, as shown here.



8. Close the schematic and layout views. Do **NOT** save.
9. Leave the Virtuoso session.
10. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you

- ◆ Used the Select Via feature for interactive routing.
- ◆ Used the middle mouse button to switch between horizontal and vertical orientations.
- ◆ Used Ctrl + Mouse Wheel to change the via offset point.



Module 4: Assisted Wires Creation

Lab 4-1 Using Pin-to-Trunk Routing

Objective: To explore the trunk composition to route a cell.

In this lab, you

- ◆ Explore how to define a trunk using an existing routing object.
- ◆ Automatically create the device-level routing.
- ◆ Route the nets using the Pin-to-Trunk router in the layout view.

Lab Setup

1. Follow the steps below to set up the environment and perform the lab:
2. If you continue from the previous lab, move to the sub-heading: [Starting the session](#). Otherwise, proceed to the next step.
3. Update the following line in the `/VLPT5_IC_25_1/.cshrc_VLPT5` file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
```

4. Save and close the file.
5. Source the updated class setup file.

```
Source .cshrc_VLPT5
```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./VLPT5_IC_25_1/M04_VLPT5_Assistedwire
```

2. To start the Cadence® Virtuoso® Studio software, enter this command in the same terminal

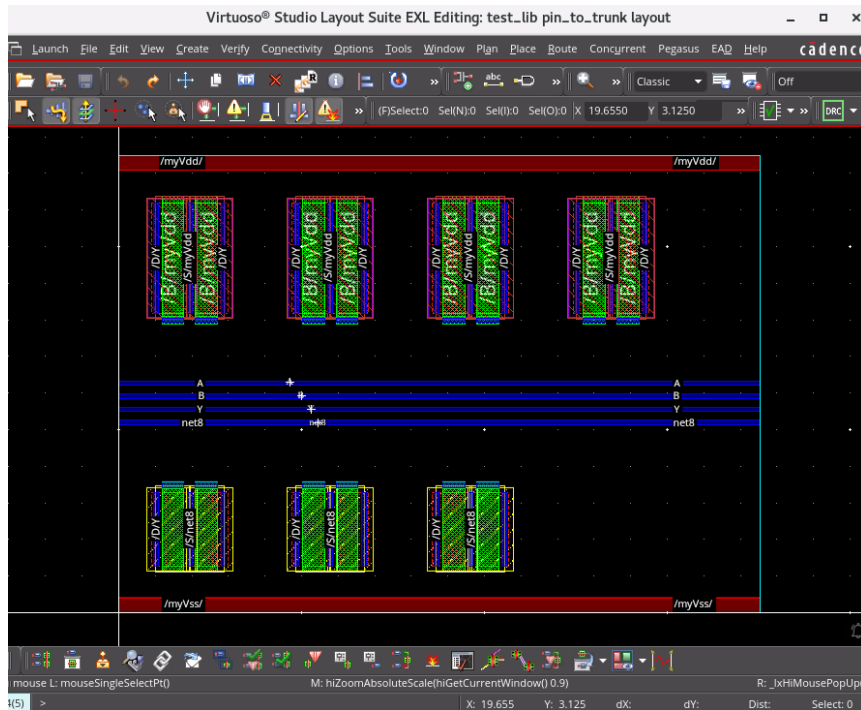
```
virtuoso &
```

3. In the CIW, choose **File – Open** and enter the following values in the form.

Library	test_lib
Cell	pin_to_trunk
View	layout

The layout view opens in XL mode.

- In the canvas, choose **Launch – Layout EXL** to open the layout in EXL mode.

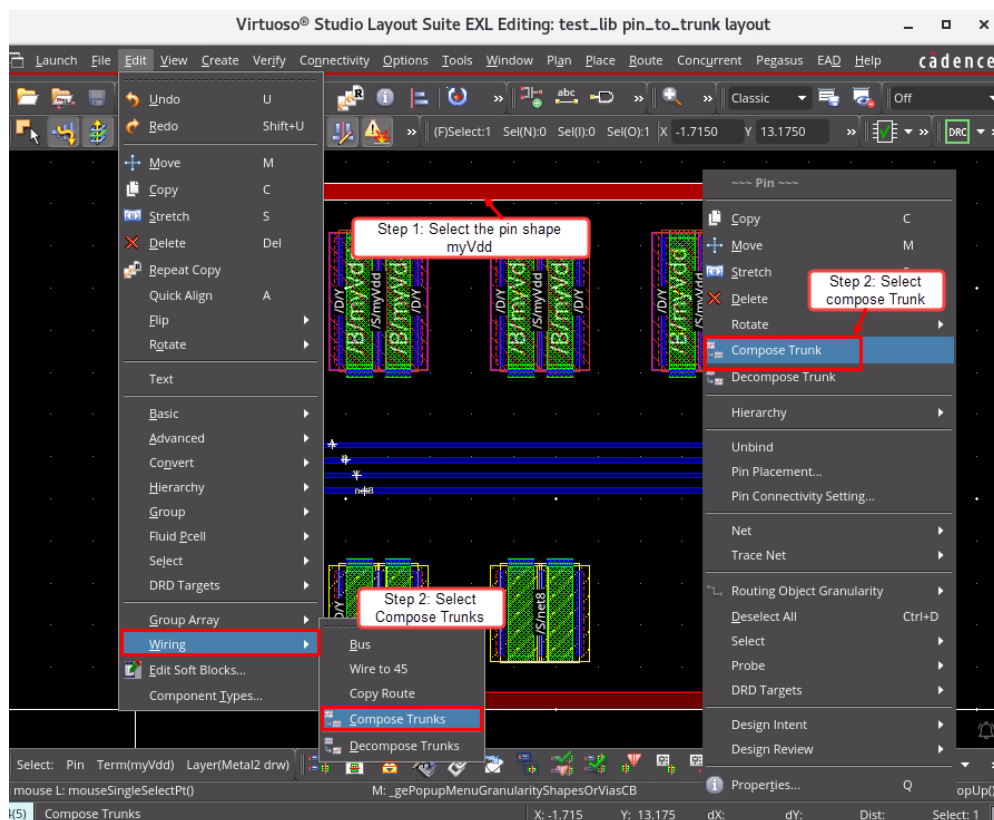


Starting Pin-to-Trunk Router with Route with Default Lookup / Route with WA Overrides

The first step is to connect the **PMOS** transistors to the **VDD power rail** (net name: **myVdd**) in the layout design.

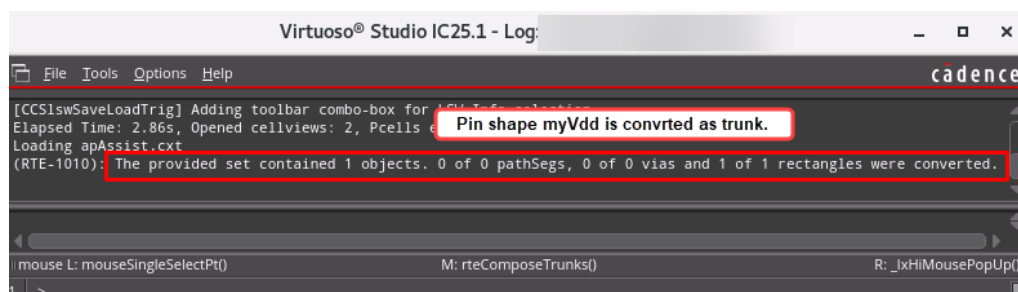
- Define a **trunk** for the power net: select the pin shape **myVdd**.

2. Choose **Edit – Wiring – Compose Trunks** or **right-click** and choose **Compose Trunk**.



The shape declared as **trunk** will be used as the **source** for all the branches.

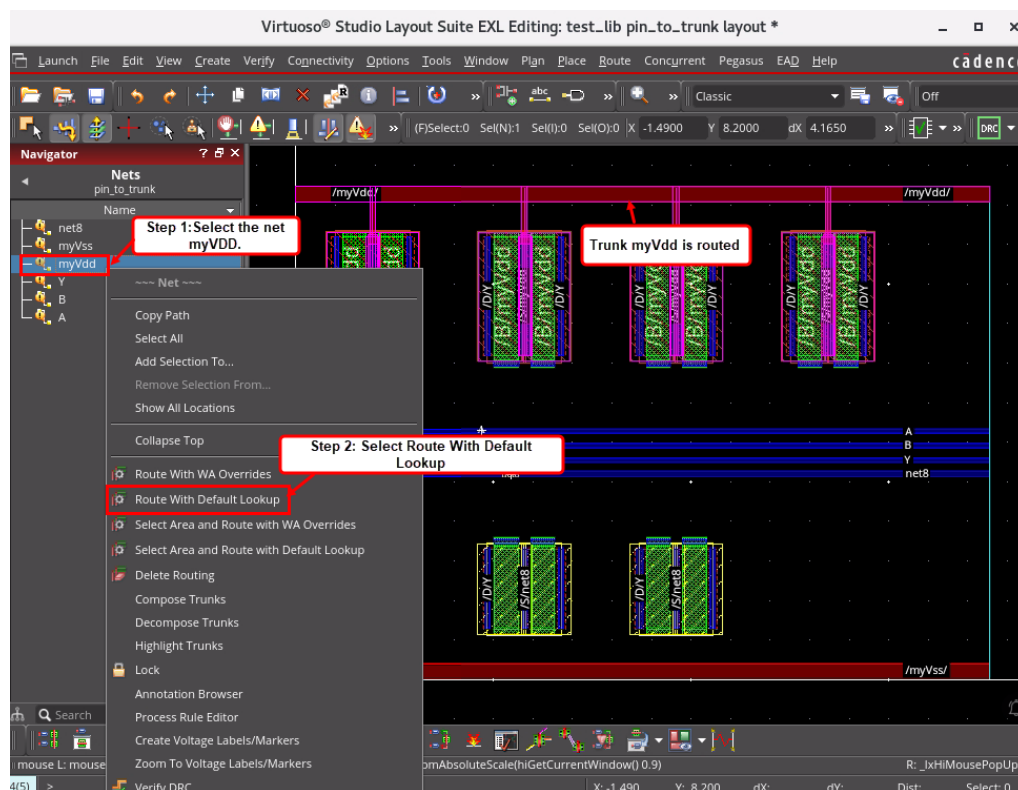
3. Observe the CIW window or *CDS.log* file for the trunk conversion information.



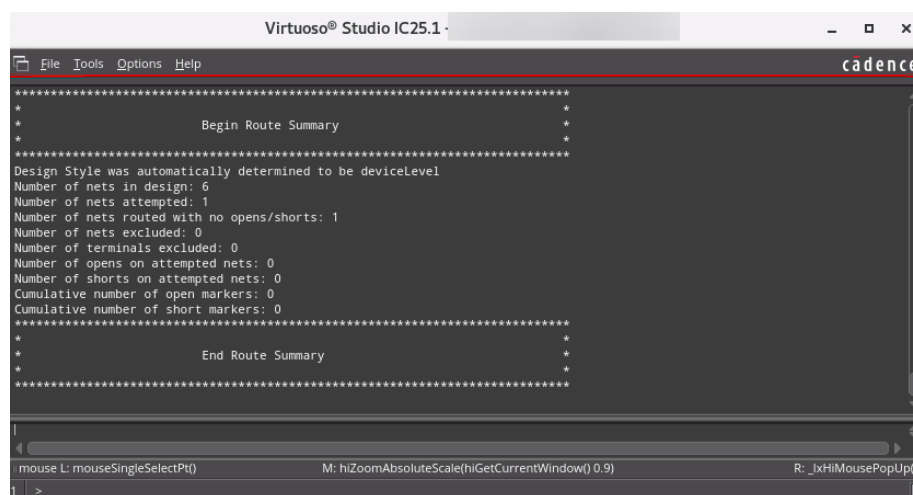
4. Choose the **Window – Assistants – Navigator** menu command and set the **OBJECTS** category in the **Navigator** to **Nets**.
5. Select the net **myVdd** in the **Navigator** assistant.
6. Start the **pin-to-trunk** router on the defined trunk **myVdd**: **Right-click** and choose **Route With Default Lookup**.

Note: In this case, there would be no difference between *Route With Default Lookup* and *Route With WA Overrides* since you have not changed anything in the *Override Constraints* section of the Wire Assistant.

Routing for the net **myVdd** has been completed using the **trunk** approach, as shown here.



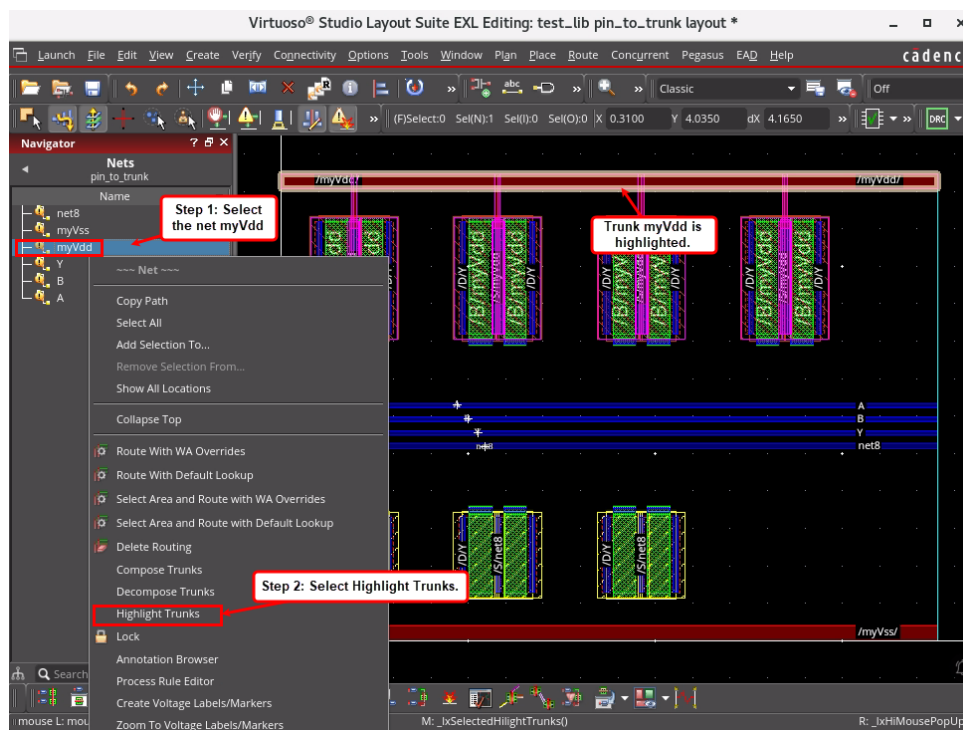
- Observe the *CDS.log* file or CIW window for the *Route Summary*.



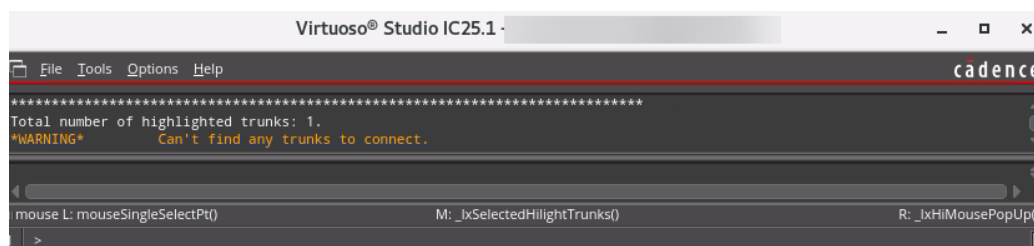
- To highlight the trunks of a selected net in the layout canvas, select the net **myVdd** in the **Navigator** assistant.

9. Right-click and choose **Highlight Trunks**.

The net **myVdd** has been routed and the corresponding trunk is highlighted.



10. Observe the *CDS.log* file or CIW window for the highlighted trunks information.



Note: The *Route With Default Lookup* or *Route With WA Overrides* commands can be used to route the nets that do not have any trunks defined. In this case, the *Minimum Spanning Tree* algorithm (*MST Route Flow*) will be used to complete the routing.

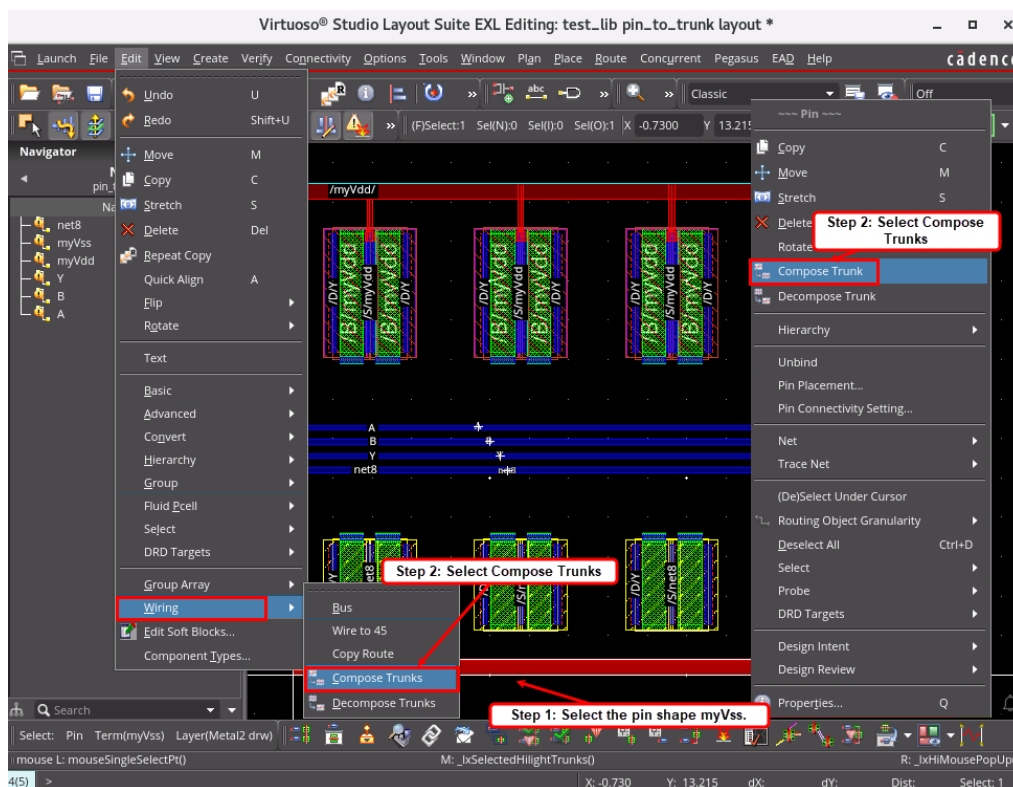
Starting Pin-to-Trunk Router with Wire Assistant

You will now explore the routing using the **Wire Assistant's Pin-to-Trunk** routing method.

The next step is to connect the **NMOS** transistors to the **VSS ground rail** (net name: **myVss**) in the layout design.

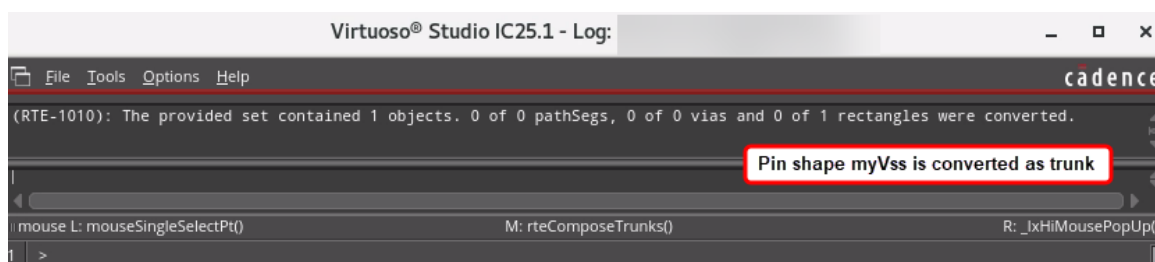
1. Define a **trunk** for the ground net: select the pin shape **myVss**.

2. Choose **Edit – Wiring – Compose Trunks** or **right-click** and choose **Compose Trunk**.



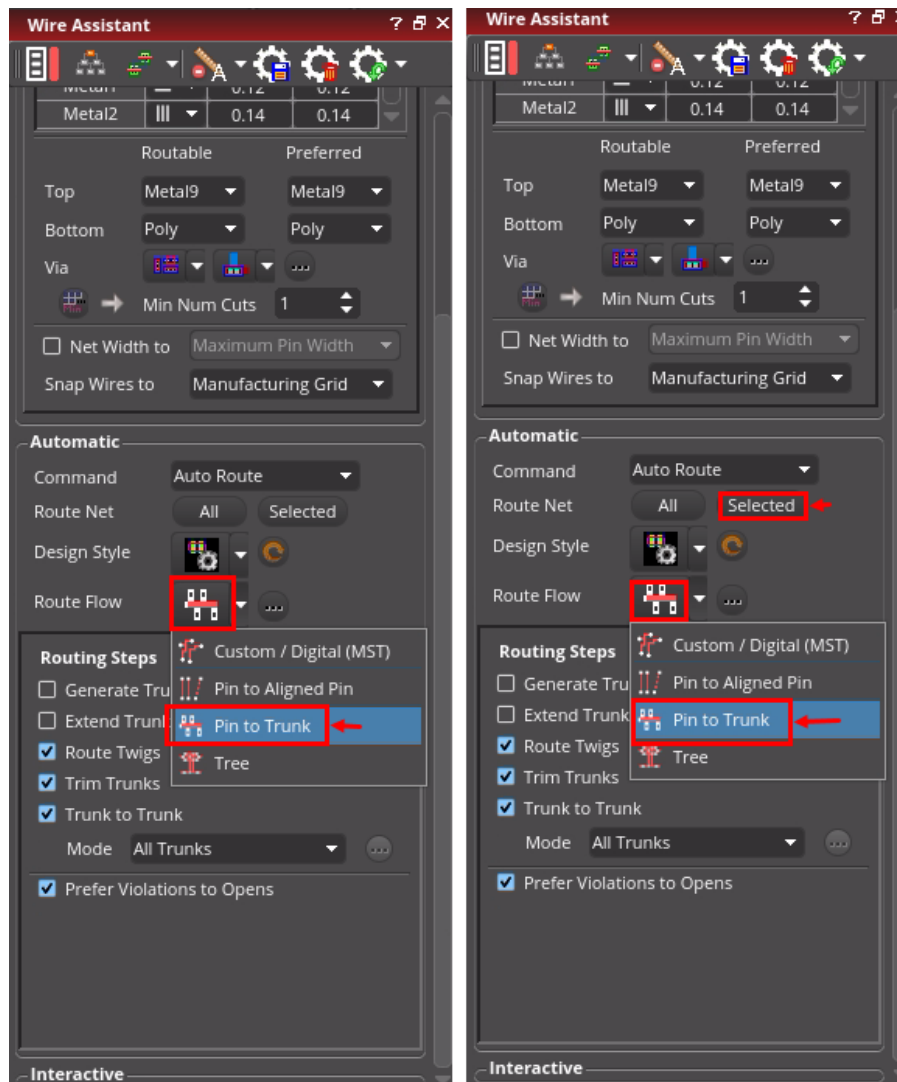
The shape declared as **trunk** will be used as the **source** for all the branches.

3. Observe the *CDS.log* file or CIW window for the trunk conversion information.

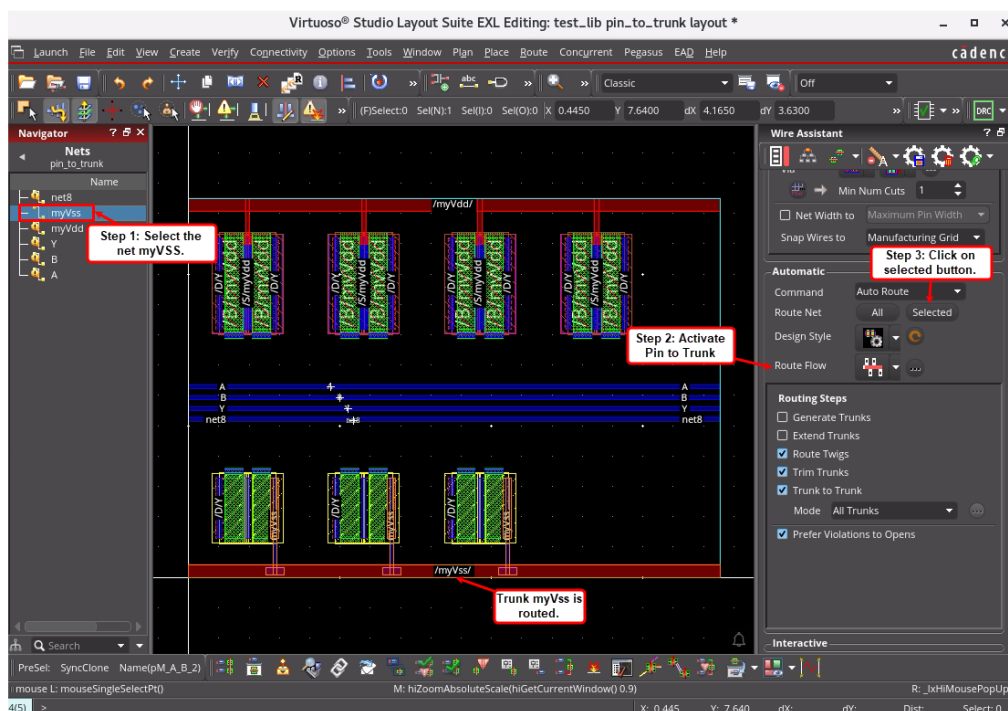


4. Activate the **Navigator** assistant if it is not already enabled in the layout design.
5. Select the net **myVss** in the **Navigator** assistant.
6. Choose the **Window – Assistants – Wire Assistant** menu command to display the Wire Assistant.
7. Choose the **Pin to Trunk Route Flow** in the **Wire Assistant**, as shown here.

8. Start the **pin-to-trunk** router on the defined trunk **myVss**, click on the **Selected** button in the **Wire Assistant**, as shown here.



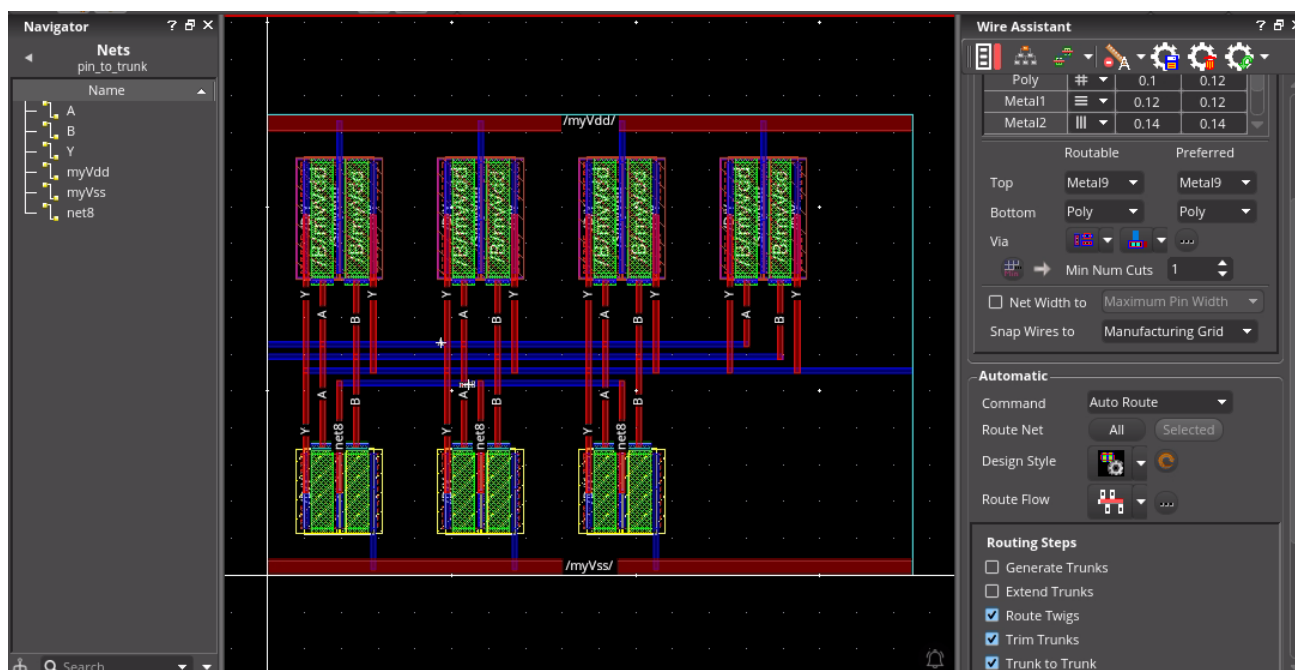
Routing for the net **myVss** has been completed using the **trunk** approach, as shown here.



- Route all the remaining nets, namely, **A**, **B**, **Y**, and **net8**: Use one of the two methods described above.

Check the quality of the results. One of the routing results is shown here.

Note: You can display the net names by using the **Options – Net Name Display** menu command.



10. Close all the cellviews without saving the changes you have made.



It is possible to create several trunks at a time by using a single “Compose Trunks” command by pre-selecting more than one trunk shape. In a similar way, several nets can be routed at the same time by selecting them in the Navigator assistant prior to the invocation of the routing command.

11. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you

- ◆ Defined a trunk using an existing routing object.
- ◆ Automatically created the device-level routing.
- ◆ Routed the nets using the pin-to-trunk router in the layout view.



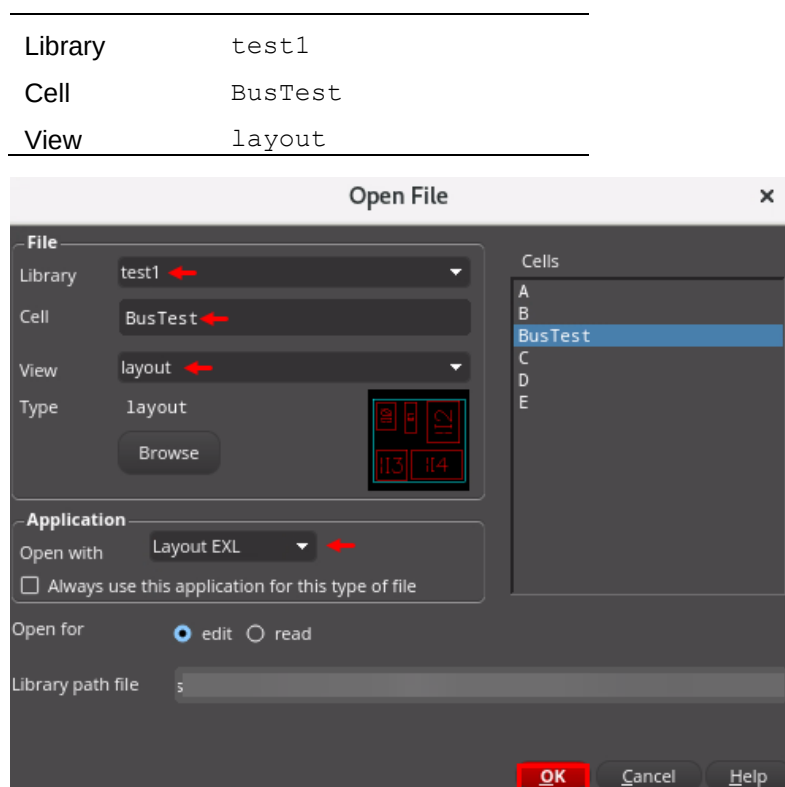
Lab 4-2 Creating Assisted Wires

Objective: To explore the assisted wires creation.

In this lab, you

- ◆ Explore the assisted commands: **Point-to-Point Routing**, **Bus Routing**, **Stranded Wiring**, and **Finish Bus** when creating wires.

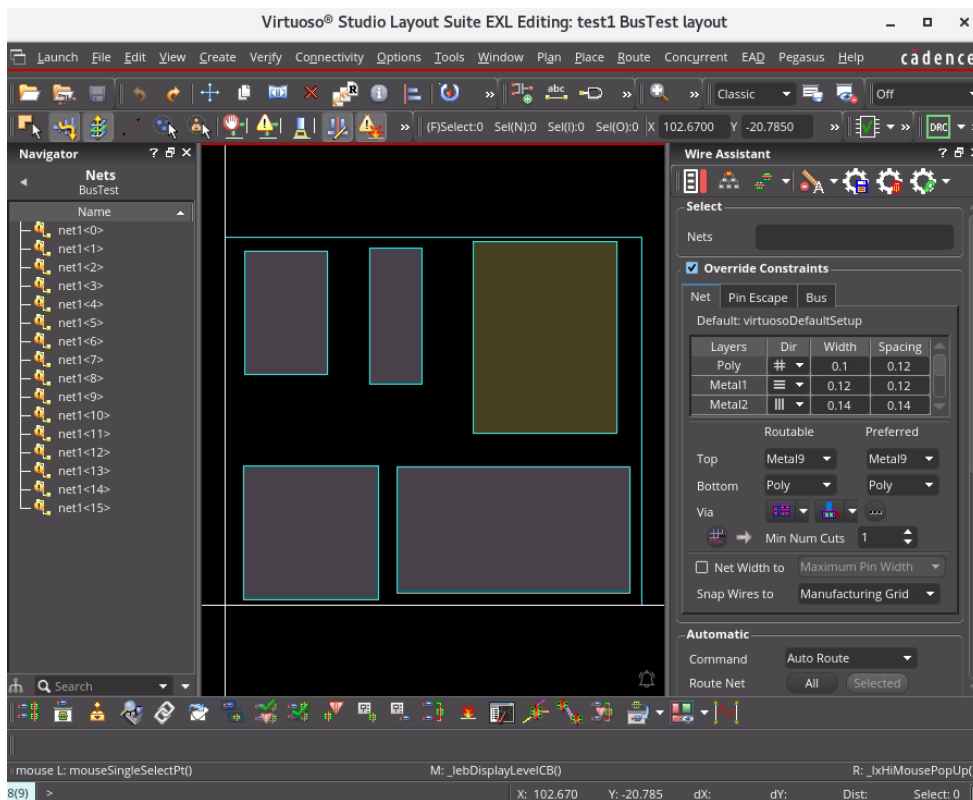
1. In the CIW, choose **File – Open** and enter the following values in the form.



The layout view opens in EXL mode.

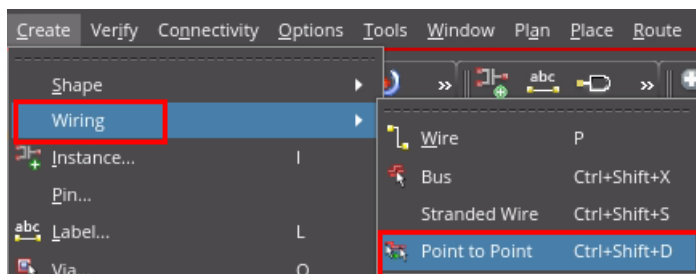
2. In the layout canvas, choose **Launch – Layout EXL** to open the layout in EXL mode.
3. In the canvas, choose the **Window – Assistants – Navigator** menu command and set the **OBJECTS** category in the **Navigator** to **Nets**.

4. In the canvas, choose the **Window – Assistants – Wire Assistant** menu command to display the Wire Assistant.



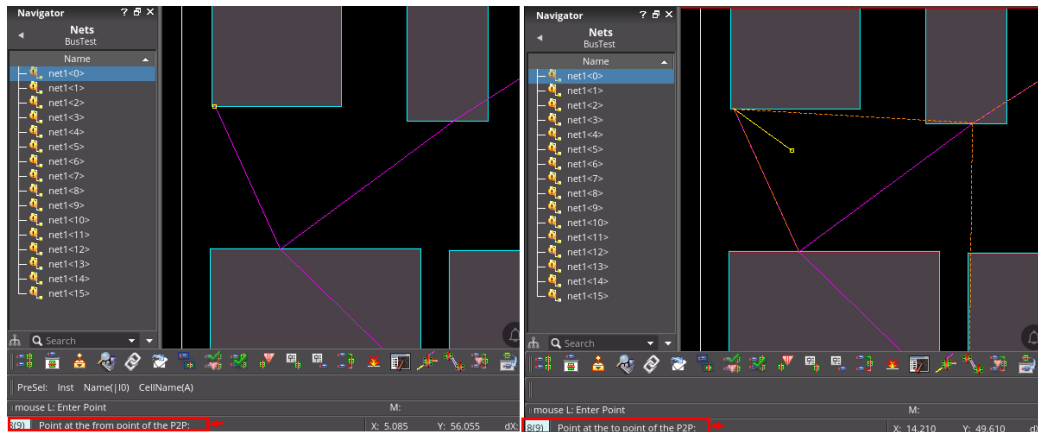
Using Point-to-Point Routing

1. In the design canvas, choose the **Create – Wiring – Point to Point** menu command or use the **Create Point to Point** icon in the Create toolbar, or use the bindkey **Ctrl+Shift+D** to invoke the Point to Point form.

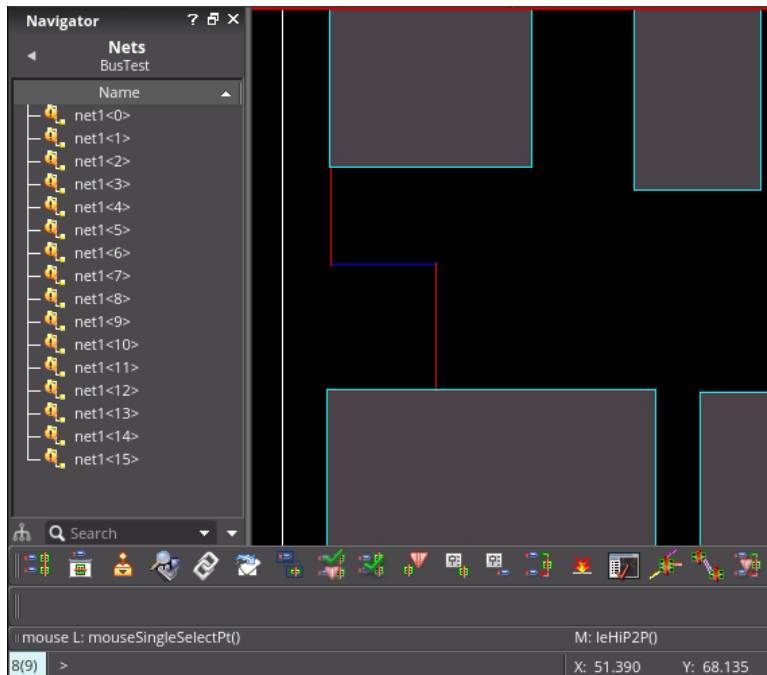


2. Select the net **net1<0>** in the **Navigator** assistant.
The pin and the flightlines highlight when you select the net **net1<0>** in the **Navigator**.

- Point at the from point of the P2P and then point at the to point of the P2P.



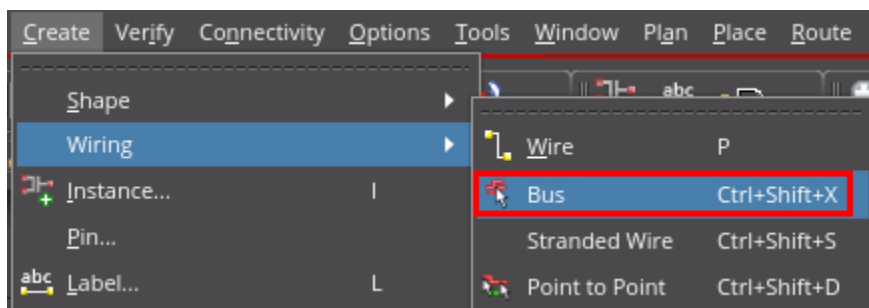
The routing is completed. Check the quality of the result. One of the routing results is shown here.



- Explore the **P2P** routing on different nets by clicking at the start point and end point and observe the routing results.
- After you have explored the **P2P** routing, choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

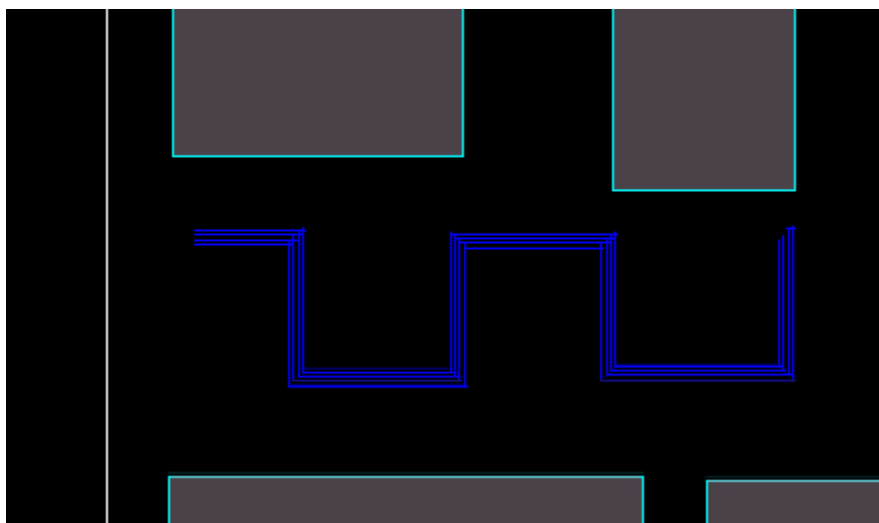
Using Bus Routing

1. In the design window, use the **Layers** Palette to select a routing layer to create the bus on, for example, **Metal1**.
2. In the design canvas, choose the **Create – Wiring – Bus** menu command or use the **Create Bus** icon in the Create toolbar or use the bindkey **Ctrl+Shift+X** to invoke the Create Bus form.



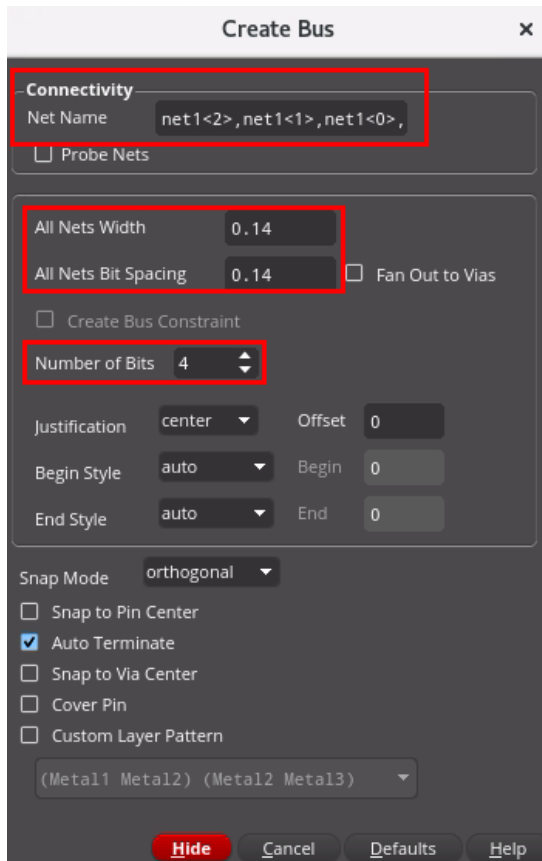
If the Create Bus form does not appear, press the **F3** key to bring up the form.

3. Set the values in the Create Bus form, as shown here.
 - a. **All Nets Width – 0.14**
 - b. **All Nets Bit Spacing – 0.14**
 - c. **Number of Bits – 4**
4. Digitize the bus and create multiple bends in free space, as shown here.



5. Press the **Esc** key to cancel the Create Bus command.

6. Choose the **Edit – Undo** menu command or click the **Undo** icon in the Edit toolbar or press the bindkey **U** to undo the created wire.
7. Display the Create Bus form again and create a bus with connectivity. In the **Net Name** field, enter **net1<0:3>**.

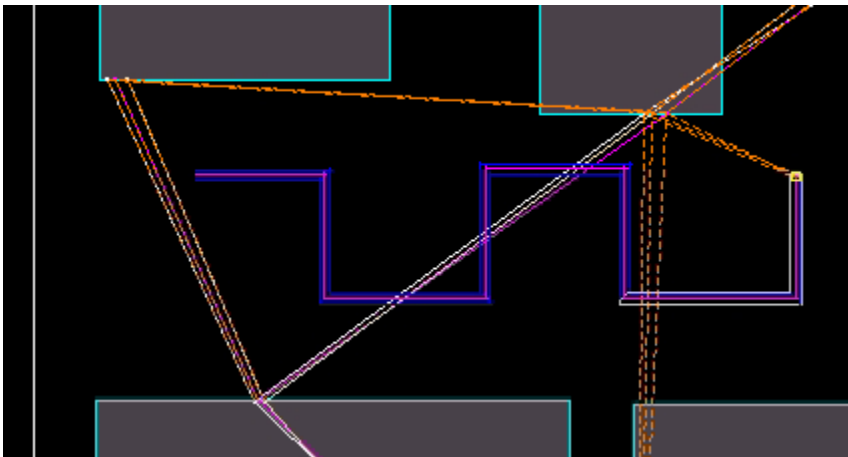


The screenshot shows the 'Create Bus' dialog box with the following settings:

- Connectivity** (highlighted):
 - Net Name: `net1<2>,net1<1>,net1<0>,`
 - ☐ Probe Nets
- All Nets Width** (highlighted): 0.14
- All Nets Bit Spacing** (highlighted): 0.14
- ☐ Fan Out to Vias
- ☐ Create Bus Constraint
- Number of Bits** (highlighted): 4
- Justification: center
- Offset: 0
- Begin Style: auto
- Begin: 0
- End Style: auto
- End: 0
- Snap Mode: orthogonal
- ☐ Snap to Pin Center
- ☒ Auto Terminate
- ☐ Snap to Via Center
- ☐ Cover Pin
- ☐ Custom Layer Pattern
- Layer Pattern: (Metal1 Metal2) (Metal2 Metal3)

Buttons at the bottom: **Hide**, **Cancel**, **Defaults**, **Help**.

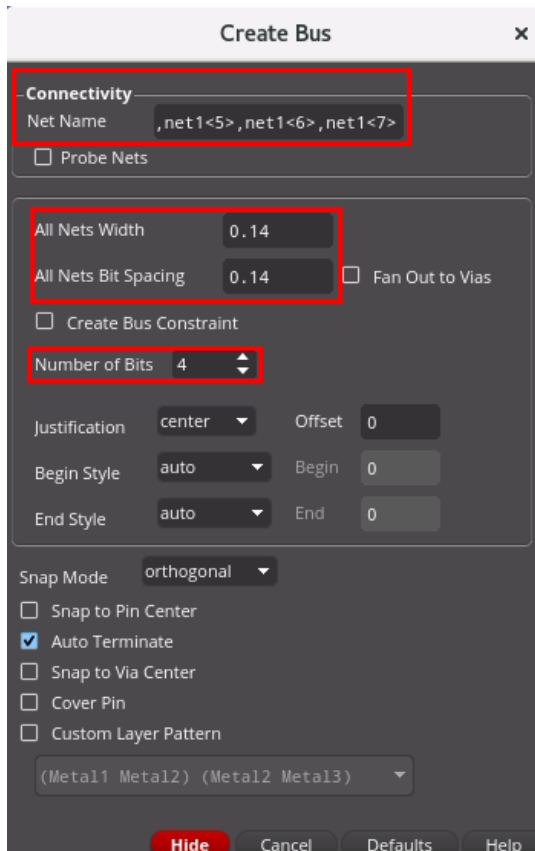
8. Digitize the bus, as shown here.



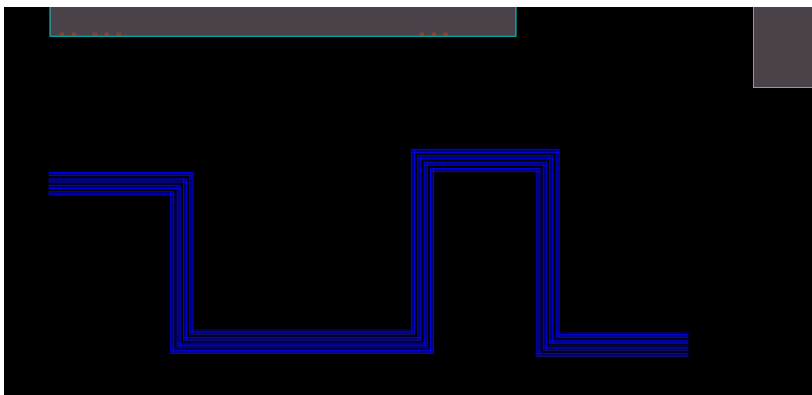
9. Cancel the Create Bus command.

10. Undo the previous edits.

11. Invoke the Create Bus form again and create a bus with connectivity. In the **Net Name** field, enter **net1<4> to net1<7>** as the members.



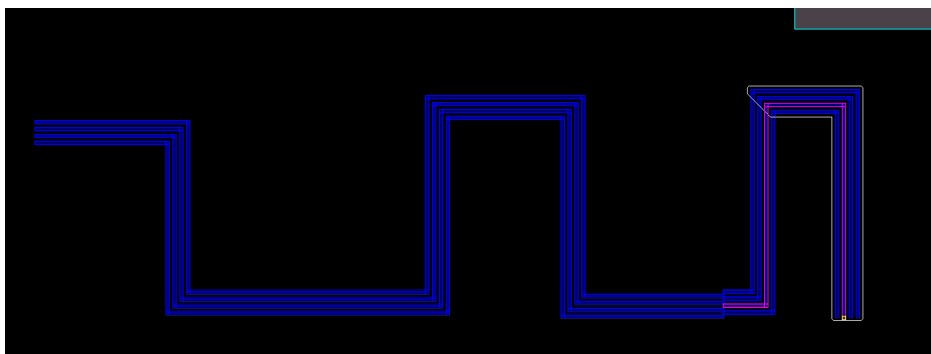
12. Digitize and create the bus, as shown here.



13. Terminate the bus.

14. Restart the bus by starting **Create Bus** and selecting only one member of the bus.

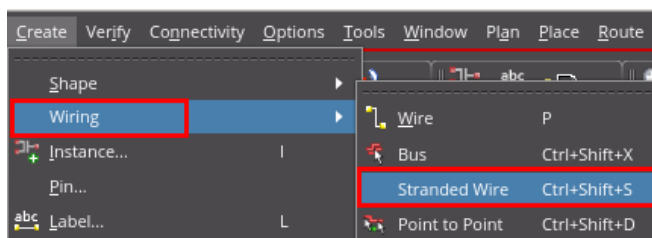
Notice that all the members of the bus will route along with the one you selected.



15. Cancel the Create Bus command.
16. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

Using Stranded Wiring

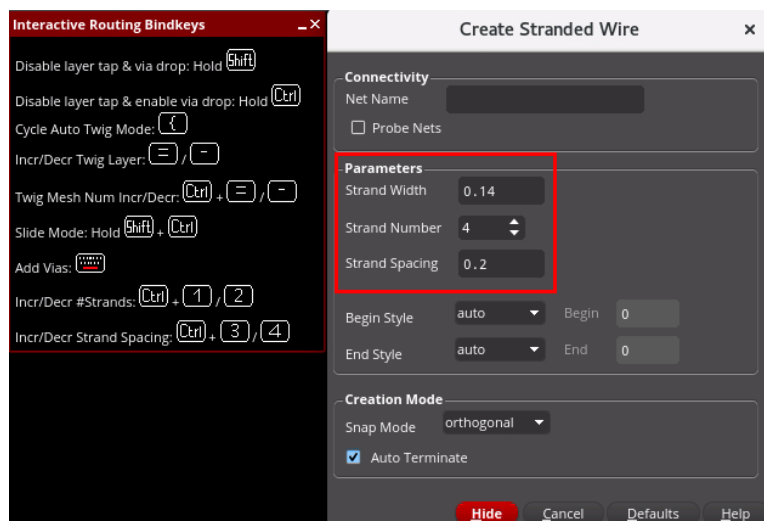
1. In the design window, from the **Layers** assistant, select a layer on which you want to create the stranded wire, for example, **Metal1**.
2. In the design window, choose the **Create – Wiring – Stranded Wire** menu command or use the bindkey **Ctrl+Shift+S** to invoke the Create Stranded Wire form.



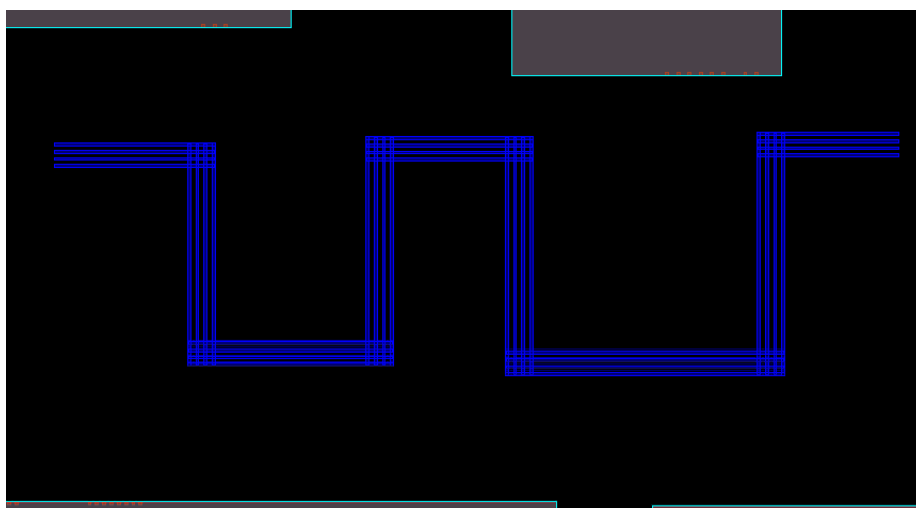
If the Create Stranded Wire form does not appear, press the **F3** key to bring up the form.

3. To create a stranded wire, set the values in the Create Stranded Wire form or use the Interactive Routing Bindkeys.

- a. **Strand Width – 0.14**
- b. **Strand Number – 4**
- c. **Strand Spacing – 0.2**

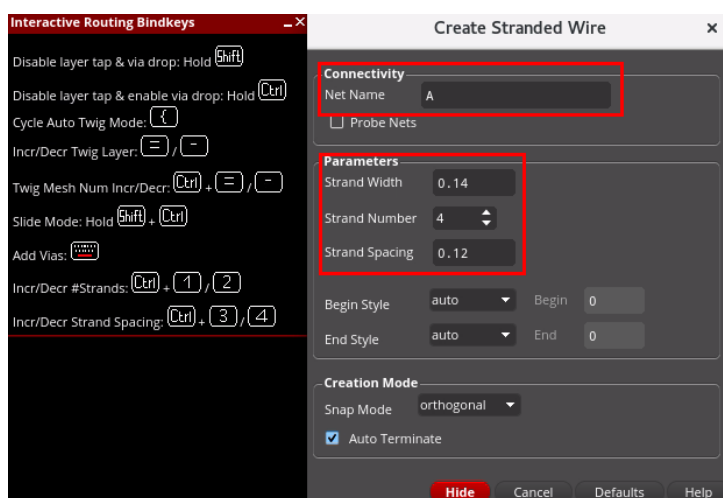


4. To create a stranded wire in an empty space in the design display area:
 - a. Click at a point in the design display area to digitize the stranded wire.
 - b. Continue to digitize the stranded wire and create bends.
 - c. Press **Enter** to finish creating the stranded wire.

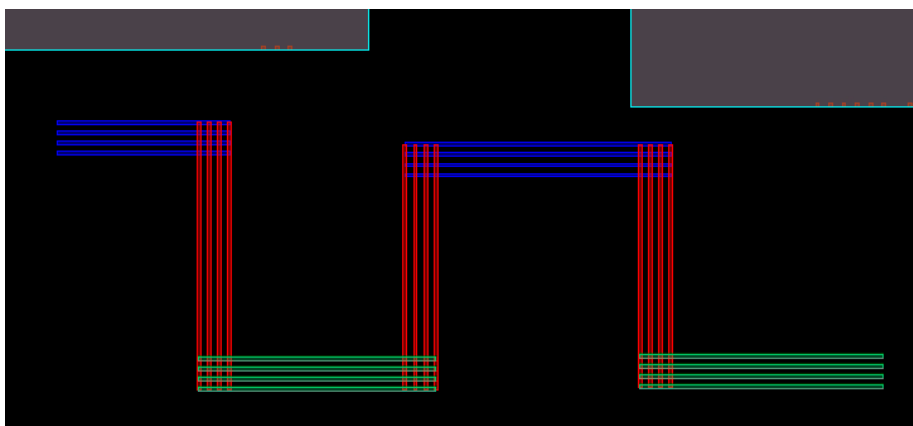


5. Press the **Esc** key to **Cancel** the Create Stranded Wire command.
6. Choose the **Edit – Undo** menu command or click the **Undo** icon in the Edit toolbar or press the bindkey **U** to undo the created wire.

7. To create a stranded wire that has connectivity, invoke the Create Stranded Wire form again, set the values in this form, or use the Interactive Routing Bindkeys. In the **Net Name** field, specify a net name for the stranded wire, for example, **A**.



8. To create a stranded wire in an empty space in the design display area:
- Click at a point in the design display area to digitize the stranded wire.
 - Continue to digitize the stranded wire and create bends.
 - Use the **Via Up** or **Via Down** options from the Create Stranded Wire context-sensitive menu to insert a via on the stranded wire.
 - Press **Enter** to finish creating the stranded wire.



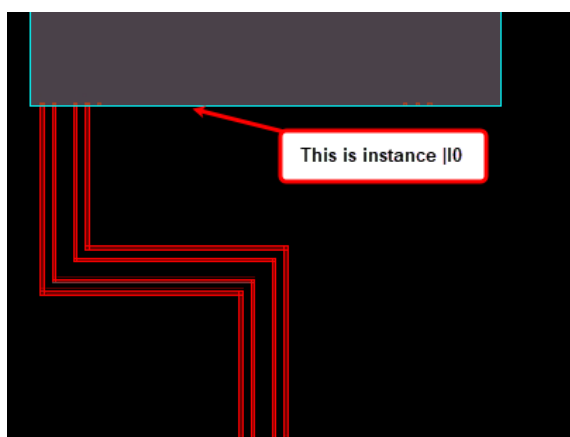
9. Cancel the Create Stranded Wire command.
10. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.

Using the Finish Bus Command

The **Finish Bus** command lets you have the remaining routing of a bus initiated by the **Create Bus** command to automatically complete, thanks to the Virtuoso® Space-Based Router engine.

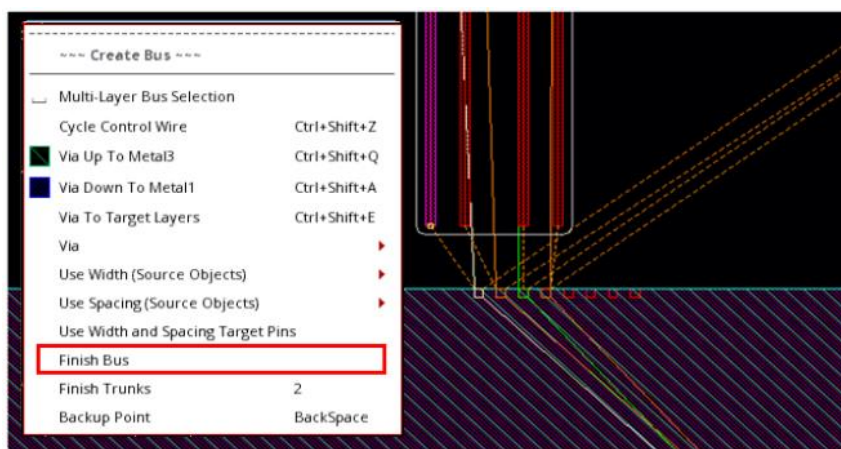
Starting Bus Routing from Instance |I0

1. In the layout view, choose the **Create – Wiring – Bus** menu command or use the **Create Bus** icon in the Create toolbar or use the bindkey **Ctrl+Shift+X** to invoke the Create Bus form and begin digitizing from the bottom left instPins of instance |I0, and leave the route incomplete, as shown here.

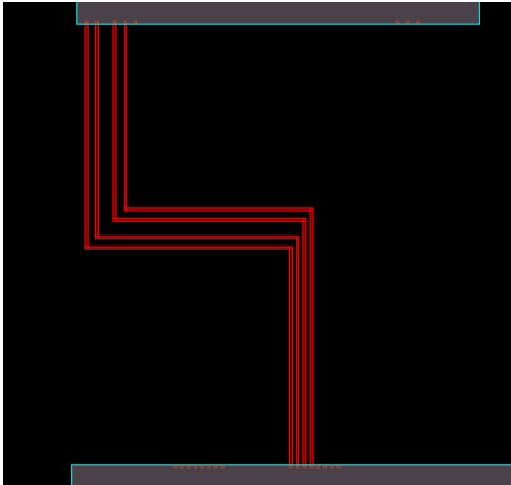


2. Start the bus routing and click on any one of the members of the incomplete bus and continue routing until you get near the vicinity of the destination pins.
3. **Right-click** and choose **Finish Bus** from the **Create Bus** menu.

Note: If you don't see the Finish Bus option, run the below command in the CIW.
`envSetVal("we" "enableFinishRoutingCmds" 'boolean t)`



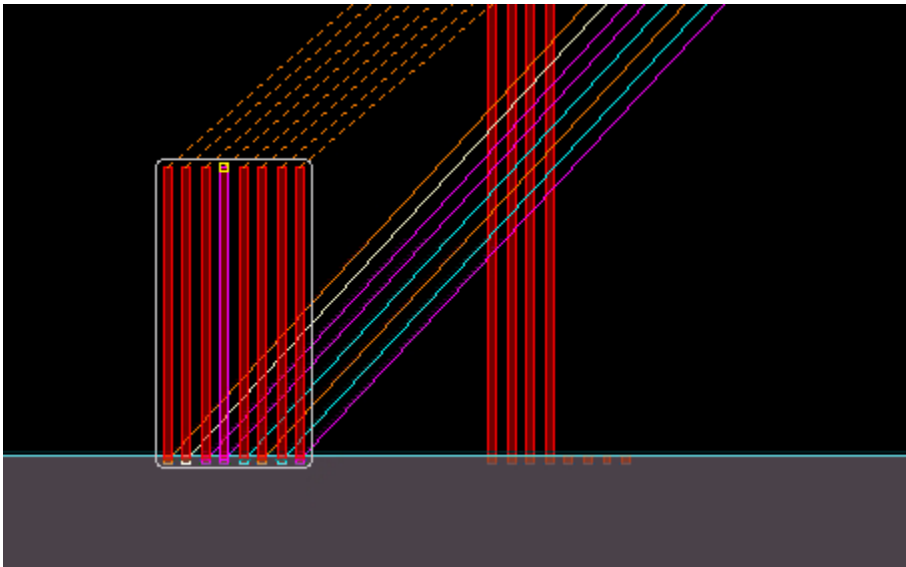
4. Verify that your results look something like this.



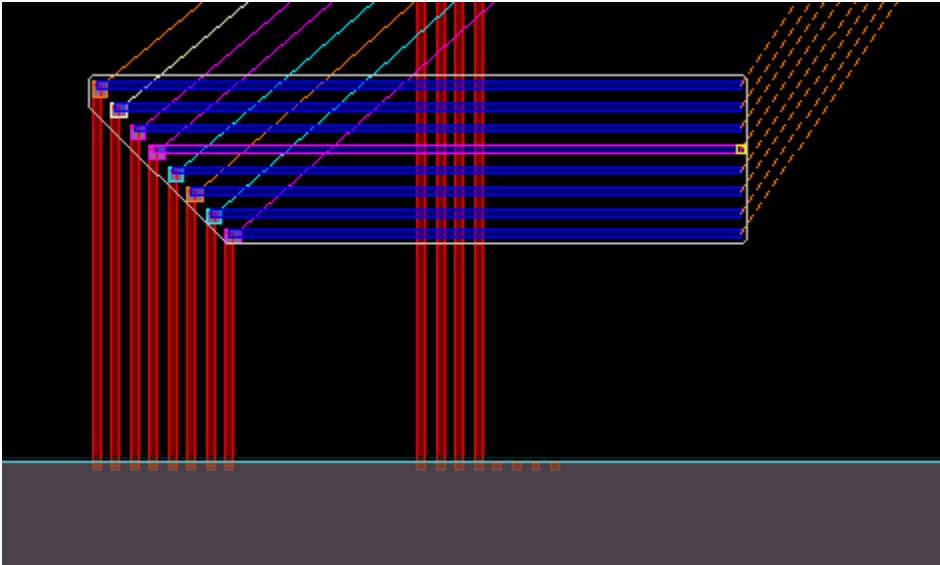
Starting Bus Routing with *Use Width and Spacing Target Pins* Option

Now, you will test to see what happens when buses are completed to pins that are split apart after the initial bus is created.

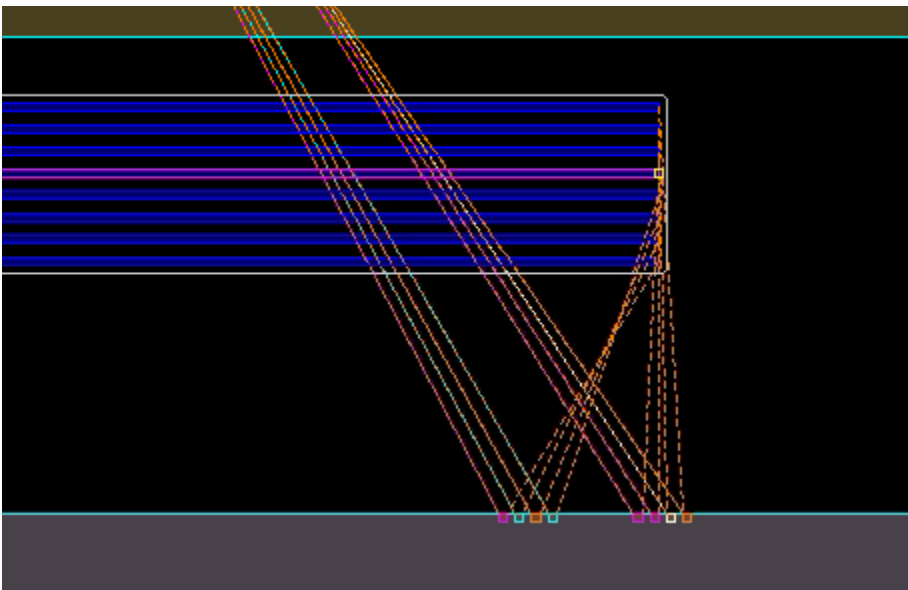
1. Start the bus routing and click on any member of the **net1<8>** to **net1<15>** bus, as shown here.



2. **Right**-click and access the Create Bus options form and choose **Via Down**. Click in the layout to add the vias and change to **Metal1**, as shown here.

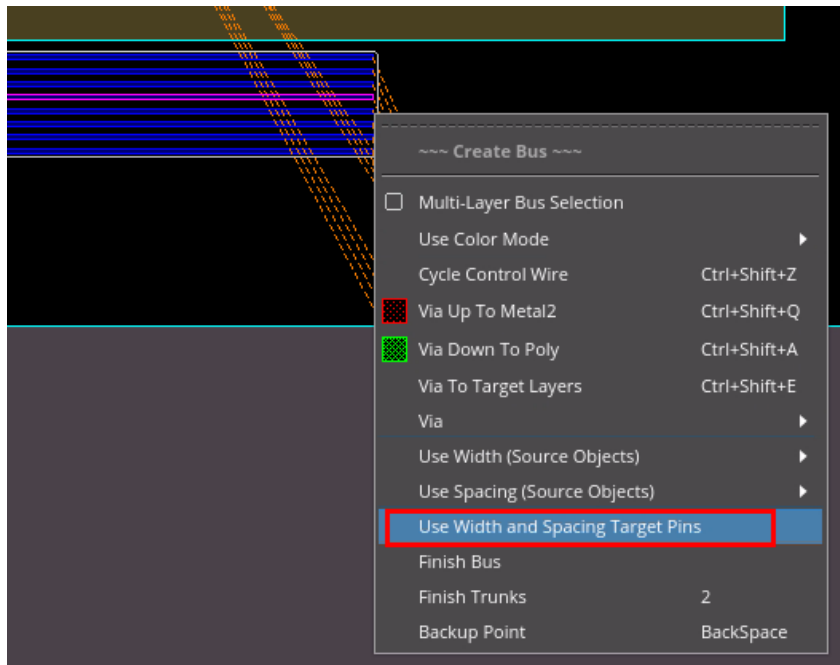


3. Continue the bus routing to the vicinity of the destination pins, as shown here.



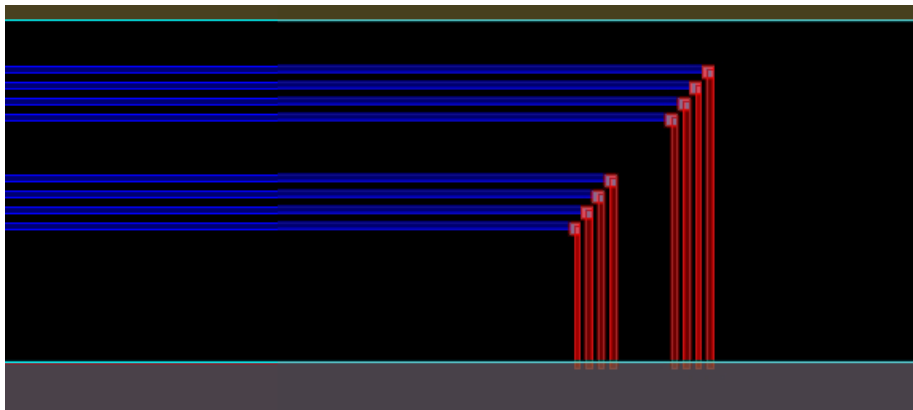
4. Again, **right**-click to access the Create Bus options form and choose **Via Up** to add the vias and change the layer to **Metal2**.

5. **Right-click** again and choose **Use Width and Spacing Target Pins**.



6. **Right-click** once more and choose **Finish Bus**.

For the clean results, try to get the flightlines as straight as possible before you execute **Finish Bus**.



7. Discard all the edits when finished testing.
8. Close all the cellviews. Do **NOT** save.
9. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you

- ◆ Used the Point to Point command
- ◆ Used the Bus Routing command
- ◆ Used the Stranded Wiring command
- ◆ Used the Finish Bus command



Lab 4-3 Using Auto Router on Selected Nets

Objective: To launch the auto router on the selected nets.

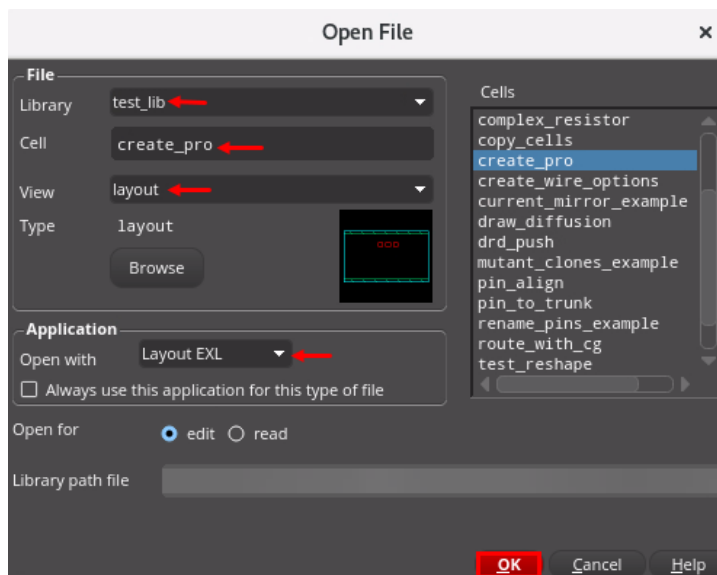
In this lab, you

- ◆ Explore how to launch the auto router on the selected nets and **Route With Default Lookup** and **Route With WA Overrides**.

The *Route With Default Lookup* or *Route With WA Overrides* commands can be used to route the nets that do not have any trunks defined. In this case, the *Minimum Spanning Tree* algorithm (*MST Route Flow*) will be used to complete the routing.

1. In the CIW, choose **File – Open** and enter the following values in the form:

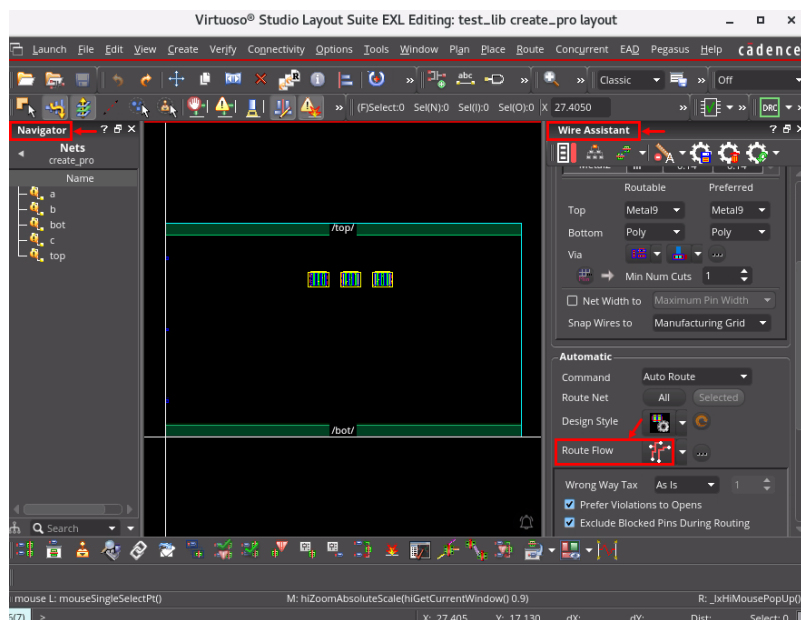
Library	test_lib
Cell	create_pro
View	layout



The layout view opens in EXL mode.

2. In the layout canvas, choose **Launch – Layout EXL** to open the layout in EXL mode.
3. In the canvas, choose the **Window – Assistants – Navigator** menu command and set the **OBJECTS** category in the **Navigator** to **Nets**.

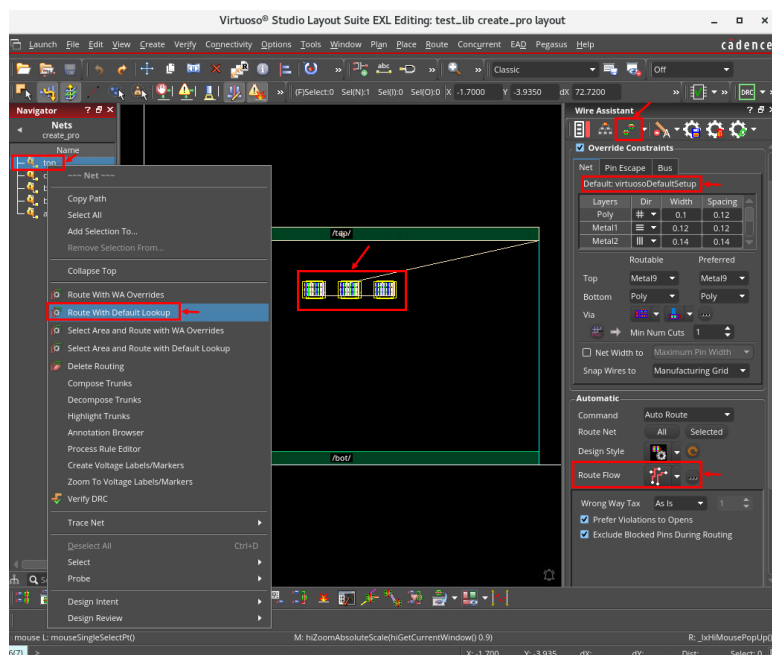
4. In the canvas, choose the **Window – Assistants – Wire Assistant** menu command to display the Wire Assistant.



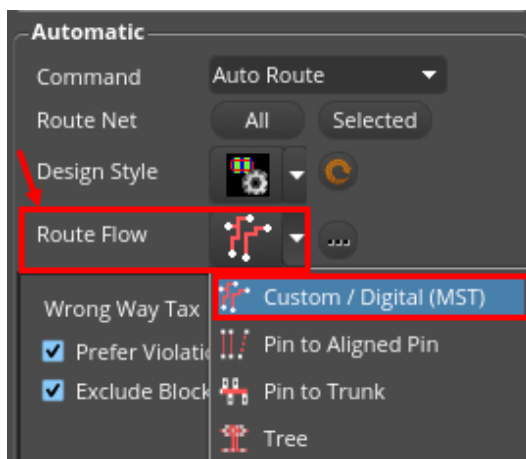
Routing with Default Lookup with Constraint Group *Default: virtuosoDefaultSetup*

You will **Route With Default Lookup**: with the net **top** and constraint group **virtuosoDefaultSetup** (default) in the layout canvas. Refer to the graphic shown here.

1. From the **Seed Attributes From a Constraint Group** drop-down list in the Wire Assistant toolbar, choose the **virtuosoDefaultSetup** constraint group.

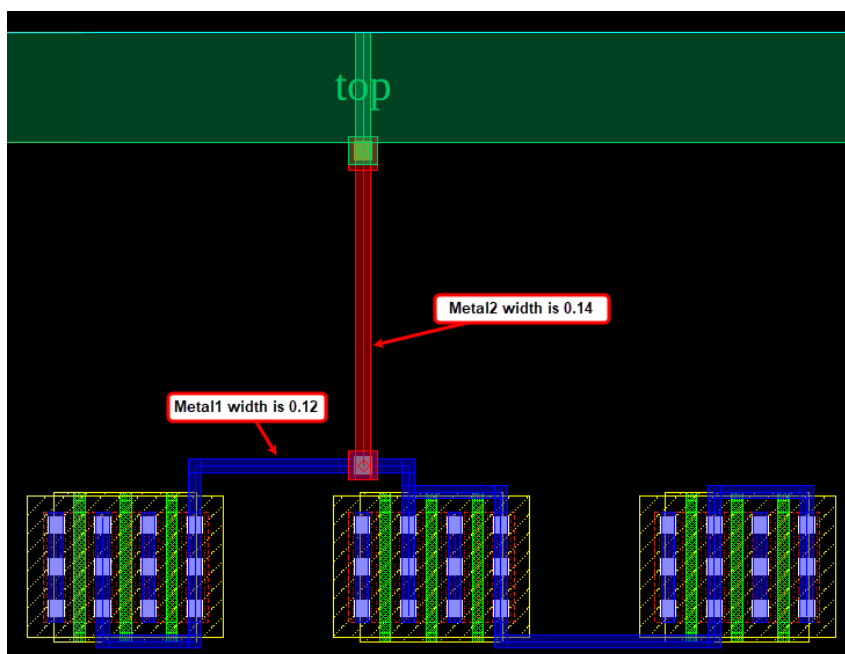


- From the *Route Flow* groupbox, choose the **Custom / Digital (MST)** option in the **Wire Assistant** toolbar, as shown here.



- Select the net **top** in the **Navigator** assistant.
- Right-click** and choose **Route With Default Lookup**.

Metal1 Width is **0.12** and **Metal2 Width** is **0.14**. The router has routed with default lookup values. Refer to the graphic shown here.

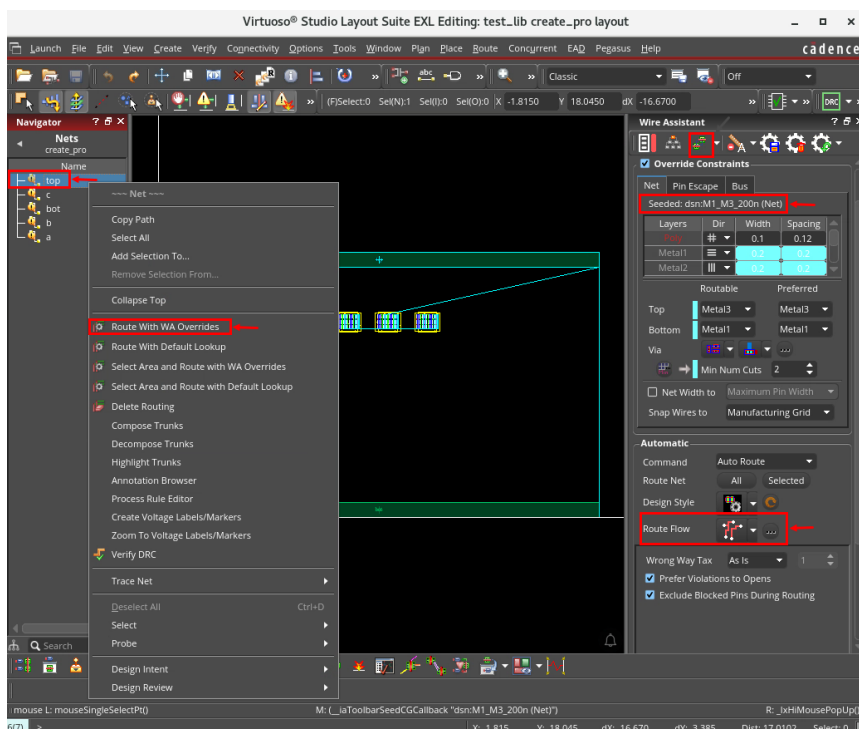


- Choose the **Edit – Undo** menu command or click the **Undo** icon in the Edit toolbar or press the bindkey **U** to undo the edits.

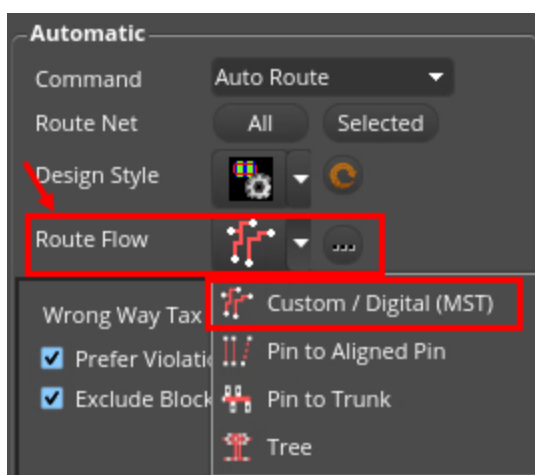
Routing with WA Overrides with Constraint Group Seeded: *dsn:M1_M3_200n (Net)*

You will **Route With WA Overrides**: with the net **top** and constraint group **dsn:M1_M3_200n (Net)** (seeded) in the layout canvas. Refer to the graphic shown here.

1. From the **Seed Attributes From a Constraint Group** drop-down list in the **Wire Assistant** toolbar, choose the **dsn:M1_M3_200n (Net)** constraint group.



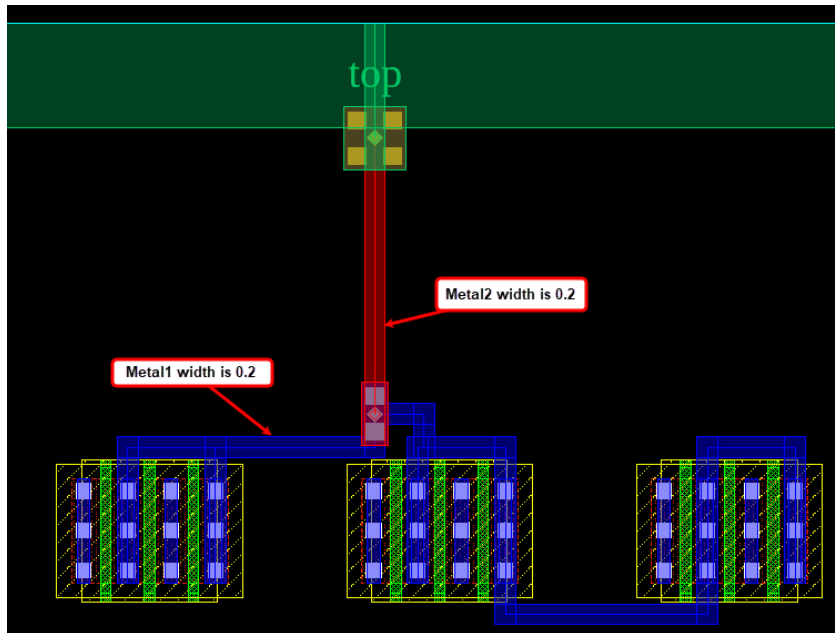
2. From the *Route Flow* groupbox, choose the **Custom / Digital (MST)** option in the **Wire Assistant** toolbar, as shown here.



3. Select the net **top** in the **Navigator** assistant.

4. **Right-click** and choose **Route With WA Overrides**.

Metal1 Width is **0.2** and **Metal2 Width** is **0.2**. The router has routed with WA Overridden (PRO) values. Refer to the graphic shown here.

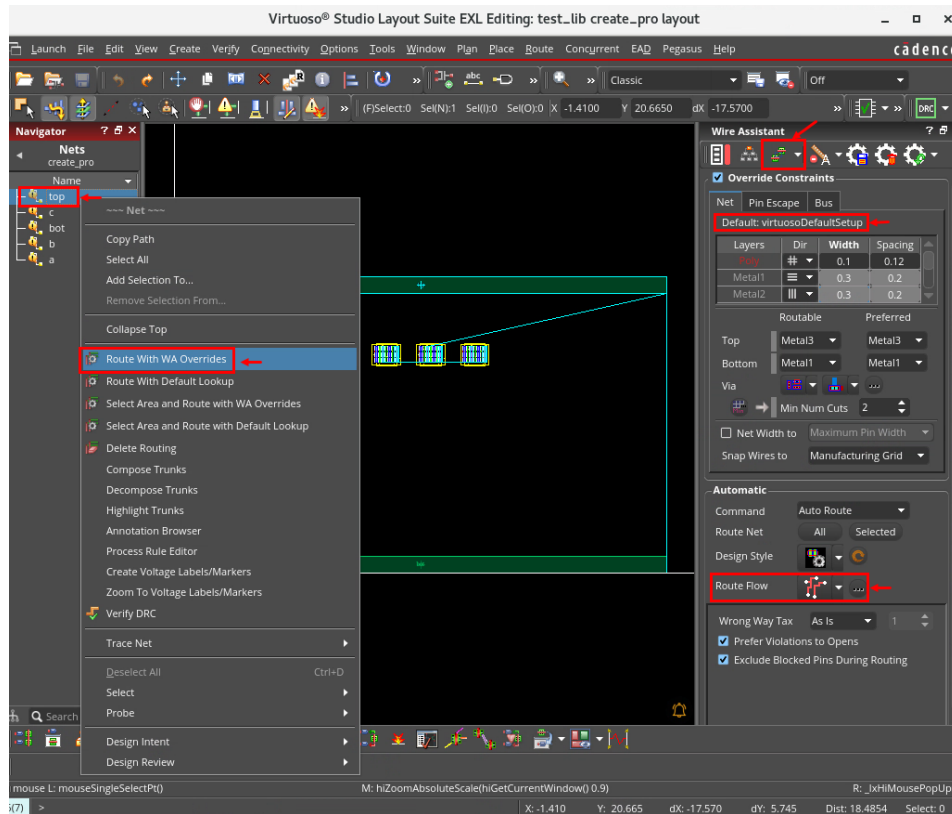


5. Undo the previous edits.

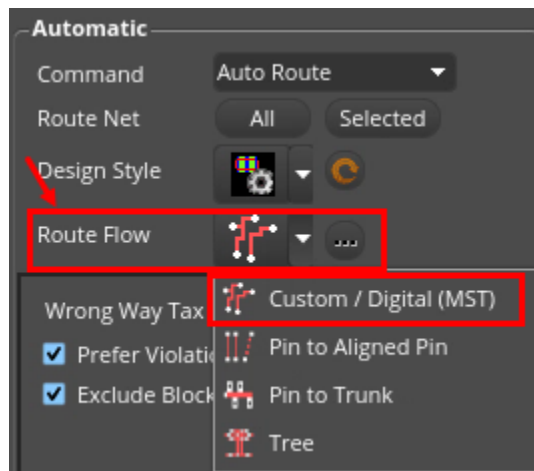
Routing with WA Overrides with *Dynamically Created PRO*

You will **Route With WA Overrides** with dynamically created PRO: with the net **top** and constraint group, which is created dynamically in the layout canvas. Refer to the graphic shown here.

1. Modify the rules in the **Override Constraints** section as in this example:
Width value of **Metal1** and **Metal2** to **0.3**.

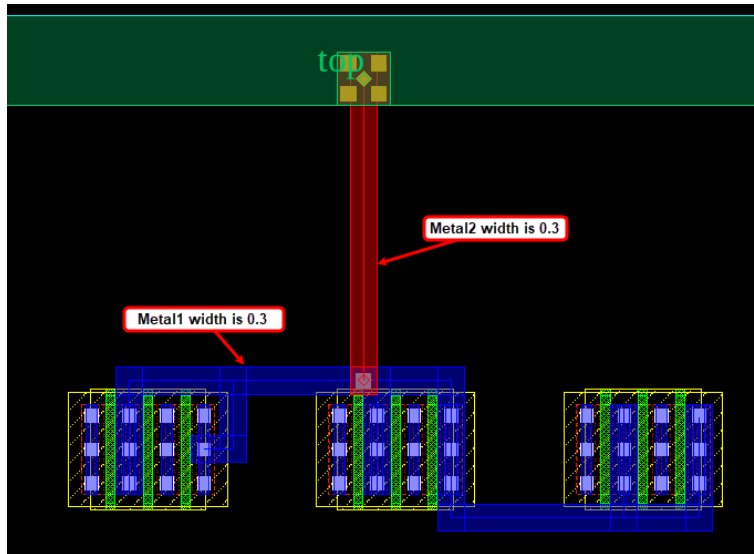


2. From the *Route Flow* groupbox, choose the **Custom / Digital (MST)** option in the **Wire Assistant** toolbar, as shown here.



3. Select the net **top** in the **Navigator** assistant.
4. **Right-click** and choose **Route With WA Overrides**.

Metal1 Width is 0.3 and **Metal2 Width is 0.3**. The router has routed with WA Overridden (PRO) values created dynamically. Refer to the graphic shown here.



5. Choose the **File – Discard Edits** menu command and click the **Yes** button in the *Discard Edits* form to discard the edits you made.
6. Close the schematic and layout cellviews without saving.
7. Leave the Virtuoso session.
8. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you

- ♦ Routed With Default Lookup with default CG
- ♦ Routed With WA Overrides with seeded CG
- ♦ Routed With WA Overrides with dynamically created PRO



Appendix A: Smart Auto Via

Lab A-1 Using Auto Via Preview

Objective: To explore the Auto Via Preview feature.

In this lab, you

- ◆ Use the **Auto Via Preview** functionality on different topologies.
- ◆ **Show Preview** enables you to see the via that you are going to create by clicking the **LMB** button.

Lab Setup

1. Follow the steps below to set up the environment and perform the lab:
2. If you continue from the previous lab, move to the sub-heading: Starting the session
Otherwise, proceed to the next step.
3. Update the following line in the `/VLPT5_IC_25_1/.cshrc_VLPT5` file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251ISR_lnx86
```

4. Save and close the file.
5. Source the updated class setup file.

```
Source .cshrc_VLPT5
```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

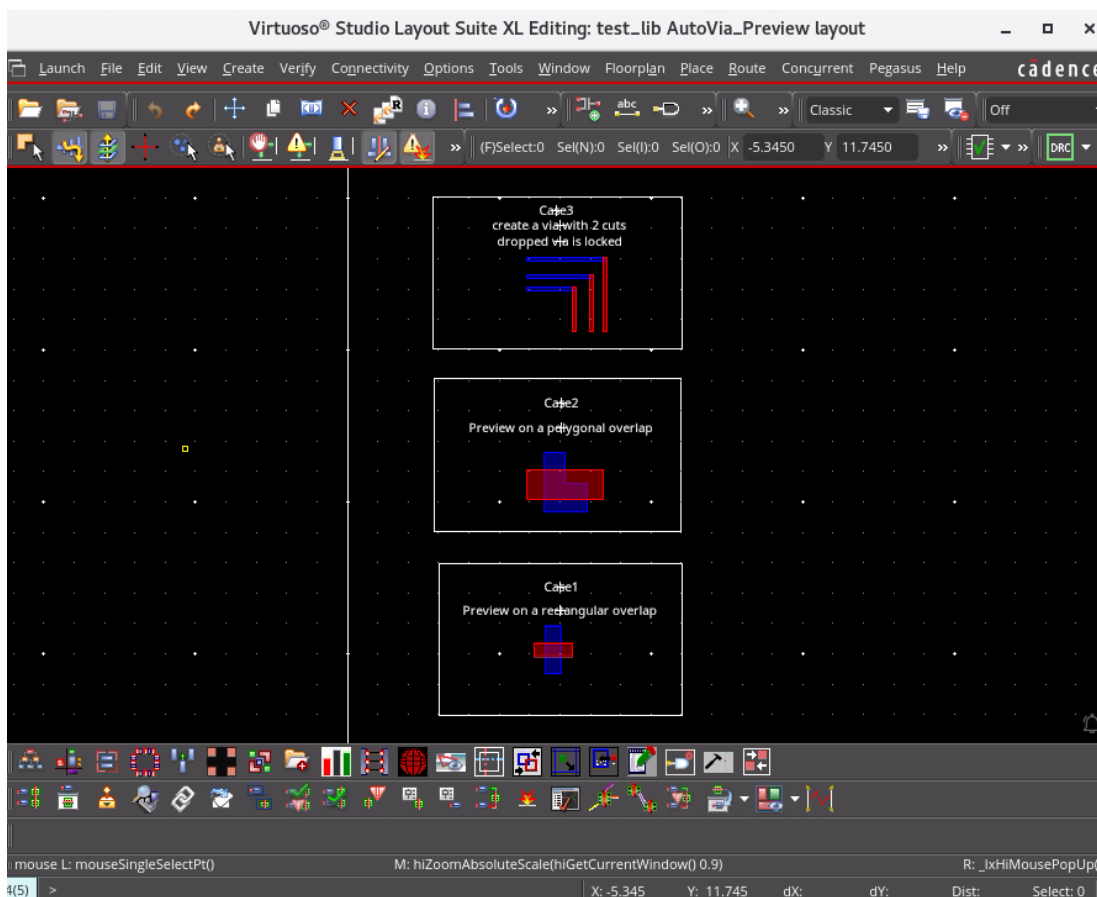
```
cd ./VLPT5_IC_25_1/M02_VLPT5_Bsc_concept
```

2. To start the Cadence® Virtuoso® Studio software, enter this command in the same terminal

```
virtuoso &
```

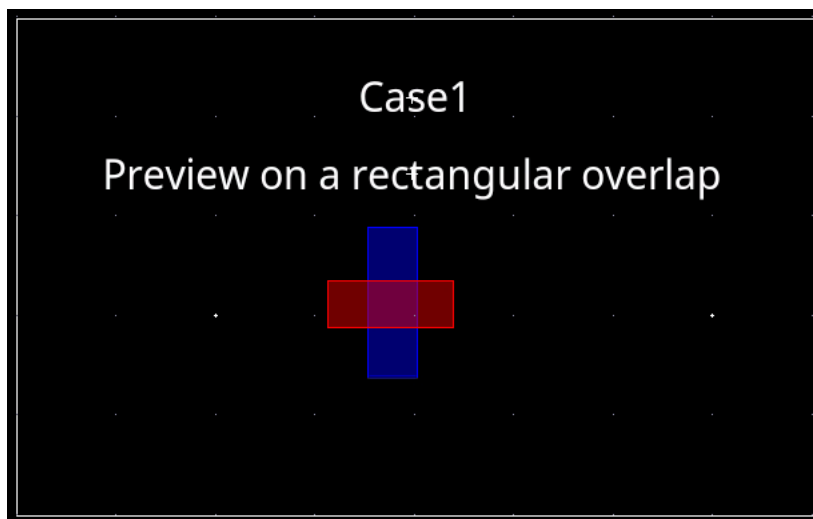
3. Open test_lib/AutoVia_Preview/layout view in EXL mode.

The Virtuoso Layout Suite EXL palette opens for editing.



Rectangular Overlap

1. In the layout canvas, choose the **View – Zoom To Area** menu command or press the Bindkey **Z** to zoom to **Case1 – Preview on a rectangular overlap**.



2. In the design canvas, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar or use the bindkey **O** to invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Auto*.
 - b. Set **Via Alignment** as *auto*.
 - c. Expand the **More Options** *Progressive Disclosure Widget*.
 - d. Select the option: **Show Preview**.
 - e. Leave the other fields as default values.

The screenshot shows the 'Create Via' dialog box with the following settings highlighted by red boxes:

- Mode**: Set to *Auto* (radio button selected).
- Via Alignment**: Set to *auto* (dropdown menu).
- Show Preview**: Checked (checkbox).

Other visible settings include:

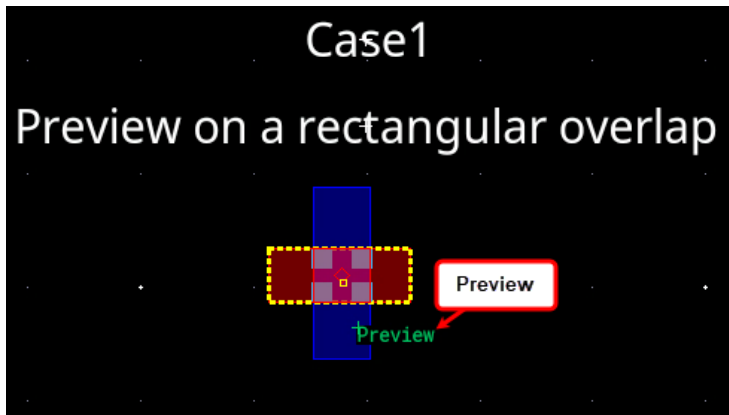
- Constraint Group**: Same as Wire
- Cut Class**: Default
- Cut Class Orientation**: auto
- Min Number of Cuts**: 1
- More Options** (expanded):
 - Align Cuts**: Checked
 - Prevent DRC Violations On The Overlap**: Checked
 - Prevent DRC Violations With Neighbors**: Unchecked
 - Prevent Via Larger Than Overlap**: Unchecked
 - Show Preview**: Checked
- Name**: via0

3. In the layout canvas, move the mouse cursor over the overlap in **Case1**.

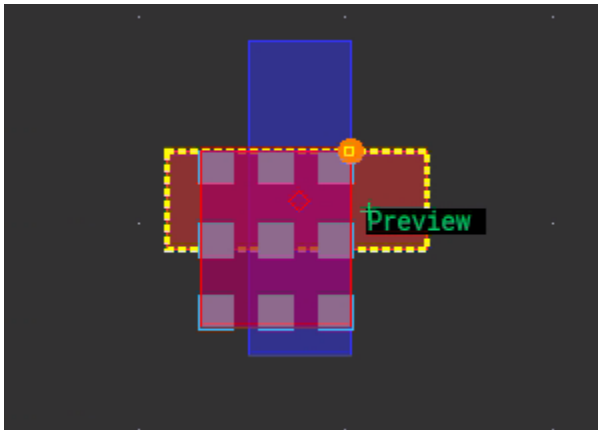
A label **Preview** is displayed close to the mouse cursor, as shown in the graphic.

For performance reasons, when the via is too complex, the application does not display it in **Preview** and a label *Preview TOO COMPLEX* is displayed instead.

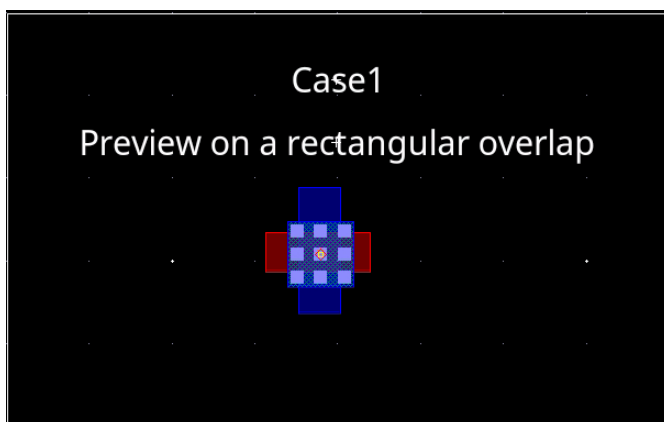
You see the via that you will drop when clicking the **LMB** button, as shown here.



4. Use the **Via Preview Bindkeys** form and increase the number of *Rows* to **3** and *Columns* to **3** in order to get a via that is bigger than the overlap.

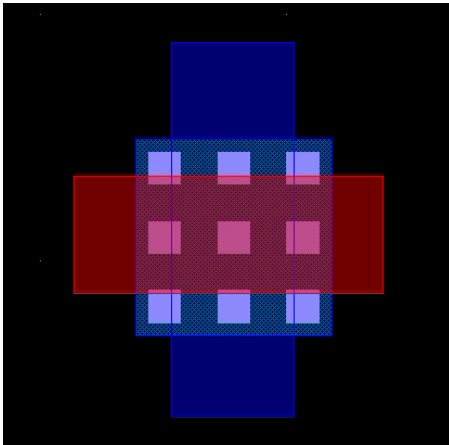


5. Do a center alignment. You can do that graphically by moving the mouse cursor. An orange circle helps you align, as shown here.



6. Click the **LMB** button in the layout canvas.

The via that you were seeing in the **Preview** is now dropped in the layout canvas.

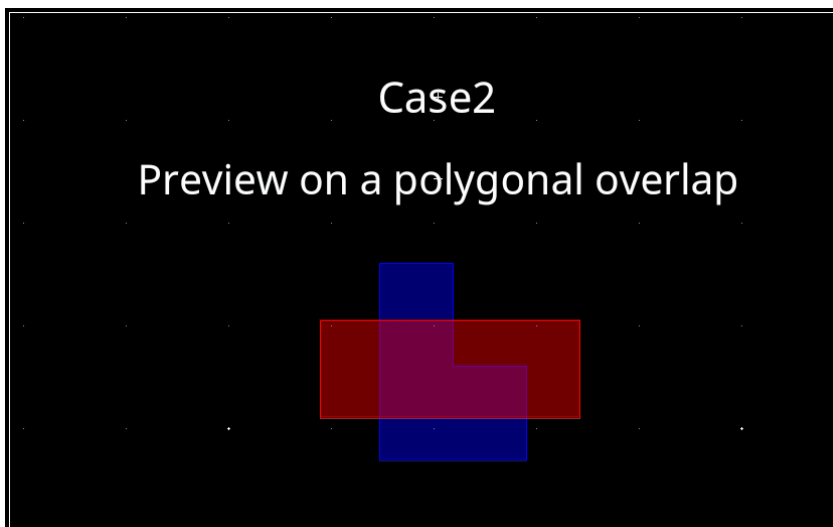


Once dropped, the label **Preview** is not displayed anymore, which helps the user to know that **Preview** mode is not enabled now.

7. Press the **Esc** key to **Cancel** the Create Via command.

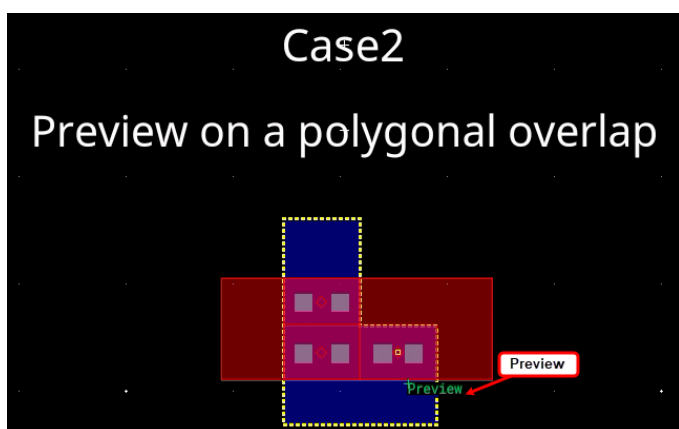
Polygonal Overlap

1. Zoom to **Case2 – Preview on a polygonal overlap**.



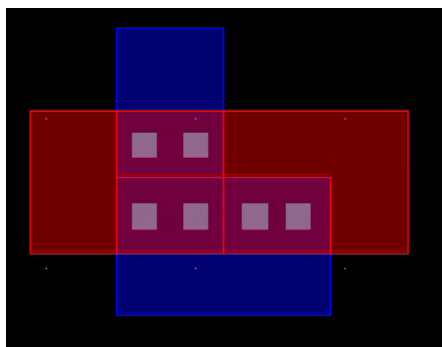
2. Invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Auto*.
 - b. Set **Via Alignment** as *auto*.
 - c. Expand the **More Options Progressive Disclosure Widget**.

- d. Select the two options: **Show Preview** if not already selected.
 - e. Leave the other fields as default values.
3. In the layout canvas, move the mouse cursor over the overlap in **Case2**.
A label **Preview** is displayed close to the mouse cursor, as shown in the graphic.
You see the via that you will drop when clicking the **LMB** button, as shown here.



The number of *Rows* and *Columns* cannot be changed when you are in a polygonal overlap because for the same transition several vias are created.

4. Click the **LMB** button in the layout canvas.
The via that you were seeing in the **Preview** is now dropped in the layout canvas.



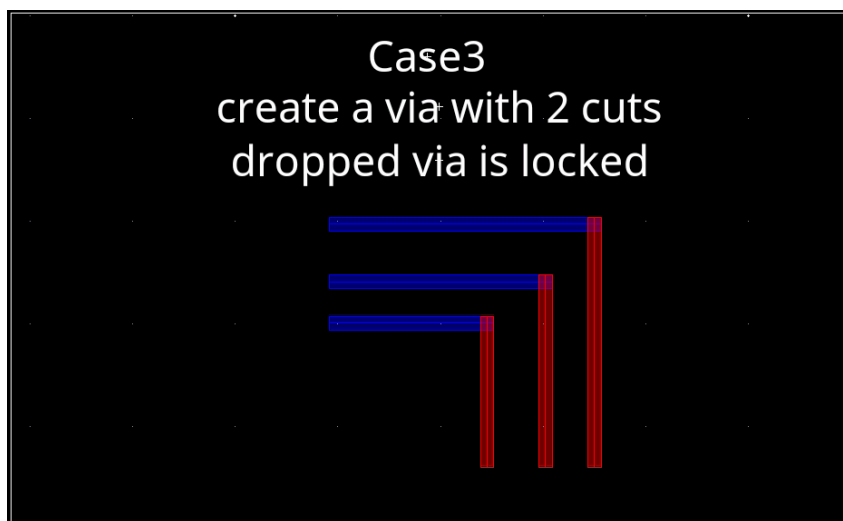
Once dropped, the label **Preview** is not displayed anymore, which helps the user to know that **Preview** mode is not enabled now.

5. Press the **Esc** key to **Cancel** the Create Via command.

Via Locked

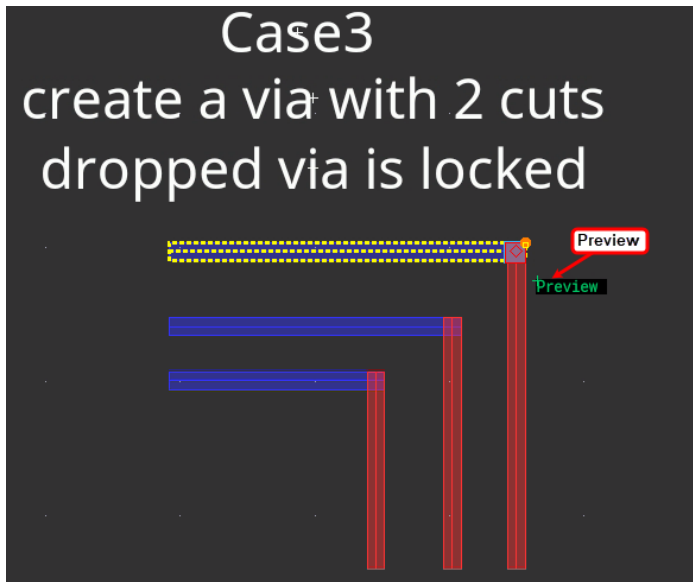
Here you see that once a customized via has been dropped, this via is locked, and you can drop it in another overlap.

1. Zoom to **Case3** – create a via with 2 cuts – dropped via is locked.

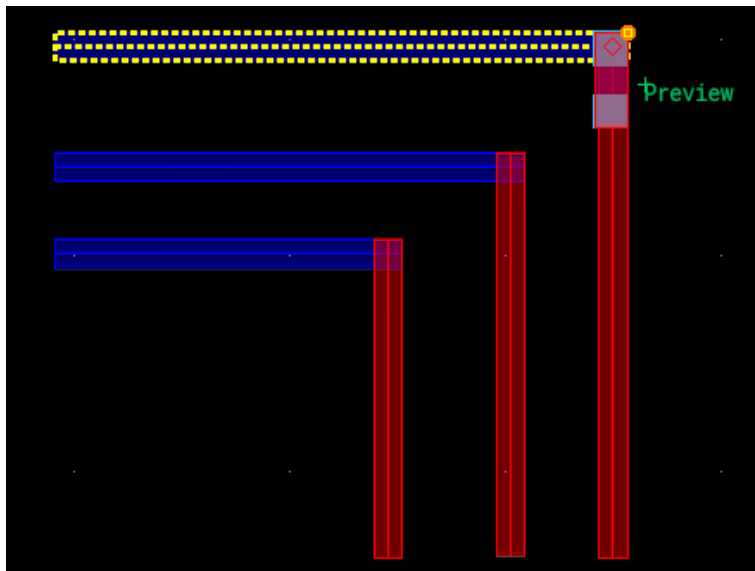


2. Invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Auto*.
 - b. Set **Via Alignment** as *auto*.
 - c. Expand the **More Options Progressive Disclosure Widget**.
 - d. Select the two options: **Show Preview** if not already selected.
 - e. Leave the other fields as default values.
3. In the layout canvas, move the mouse cursor over an overlap in **Case3**.
A label **Preview** is displayed close to the mouse cursor, as shown in the graphic.

You see the via that you will drop when clicking the **LMB** button, as shown here.

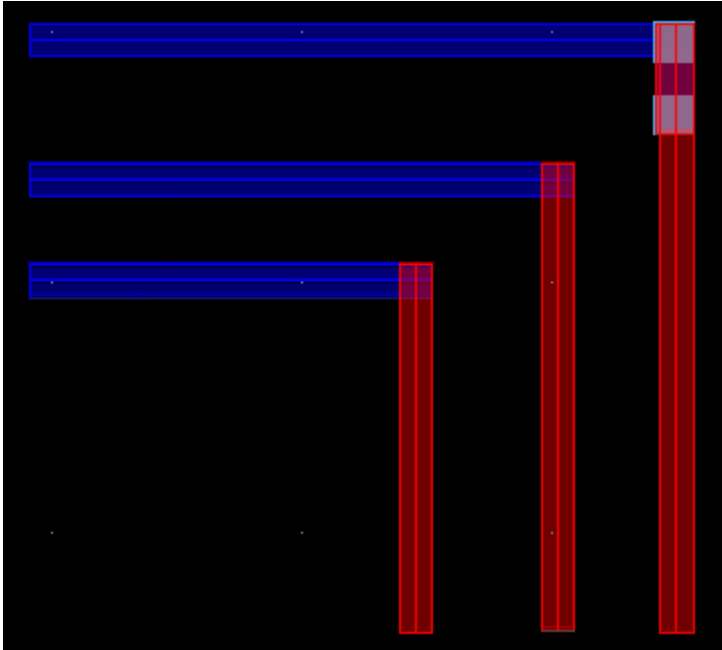


4. In the **Via Preview Bindkeys** form, use the **Ctrl + Wheel up** bindkey and increase the number of *Rows* to 2 to get a **2x1** via and move the mouse to get a top-right alignment.



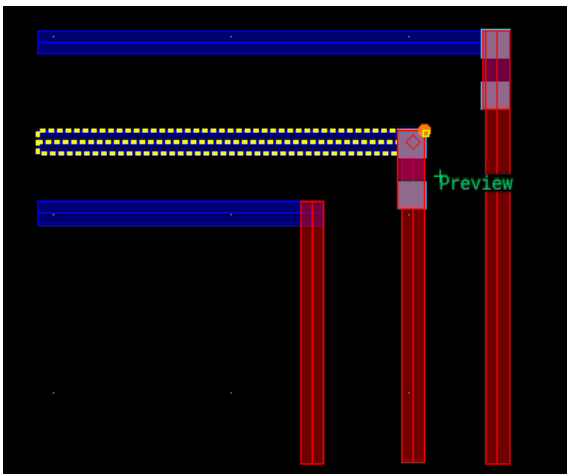
5. Click the **LMB** button in the layout canvas.

The via that you were seeing in the **Preview** is now dropped in the layout canvas.

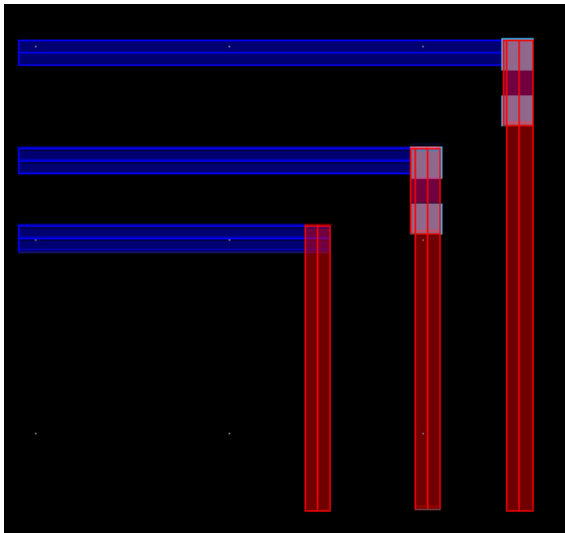


Once dropped, the label **Preview** is not displayed anymore, which helps the user to know that **Preview** mode is not enabled now.

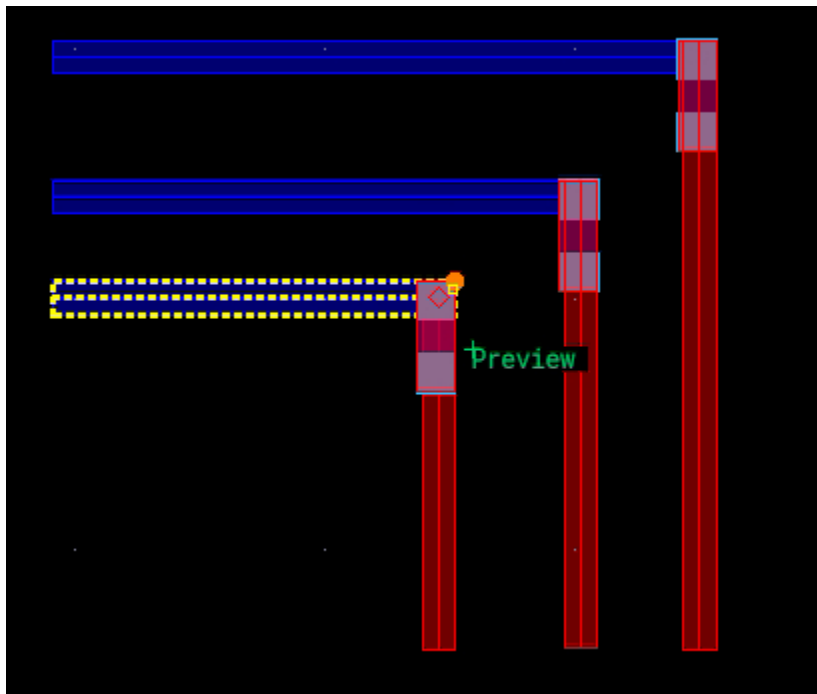
6. Move the mouse over the second overlap. The previous via is locked, and you can drop it on the second overlap.



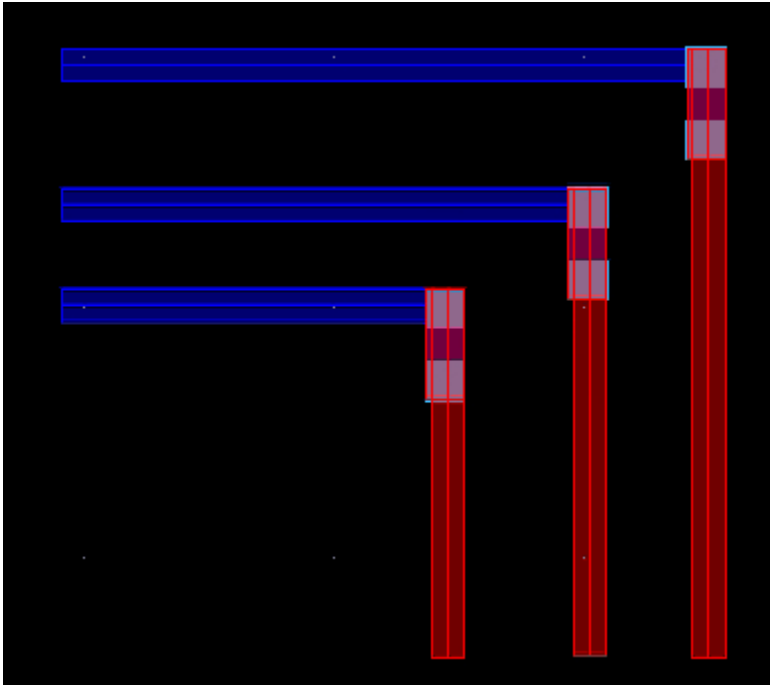
- Click the **LMB** button to drop the via in the layout canvas, as shown here.



- Repeat **Step 6** over the third overlap to get the **Preview**, as shown here.



9. Repeat **Step 7** over the third overlap to drop the via in the layout canvas, as shown here.



10. Press the **Esc** key to Cancel the Create Via command.
11. Choose the **File – Discard Edits** menu command and click the **Yes** button in the Discard Edits form to discard the edits you made.
12. Close the layout cellview without saving.
13. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you used the Auto Via Preview functionality on

- ◆ Rectangular Overlap
- ◆ Polygonal Overlap
- ◆ Via Locked



Lab A-2 Using Fast Interactive Via Editing – FIVE

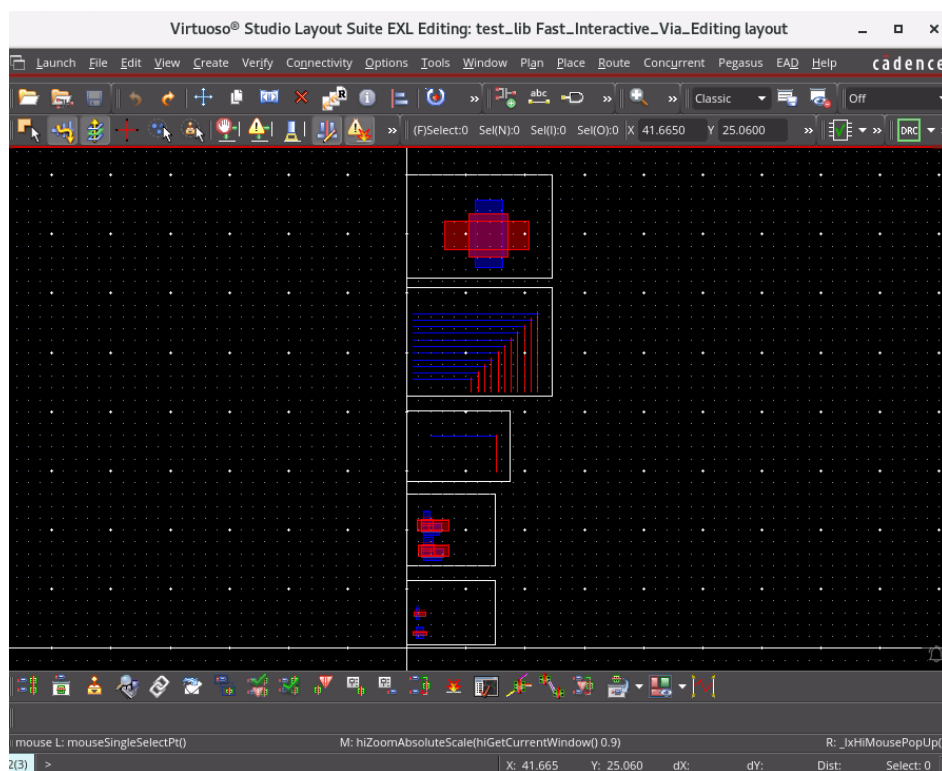
Objective: To explore the Fast Interactive Via Editing.

In this lab, you

- ◆ Learn how to use the **Fast Interactive Via Editing (FIVE)** to edit/modify the existing vias when the overlapping area increases/decreases to match the new topology.
- ◆ **Fast Interactive Via Editing** is also very handy in case the user needs to edit/modify the existing vias after a techfile update.

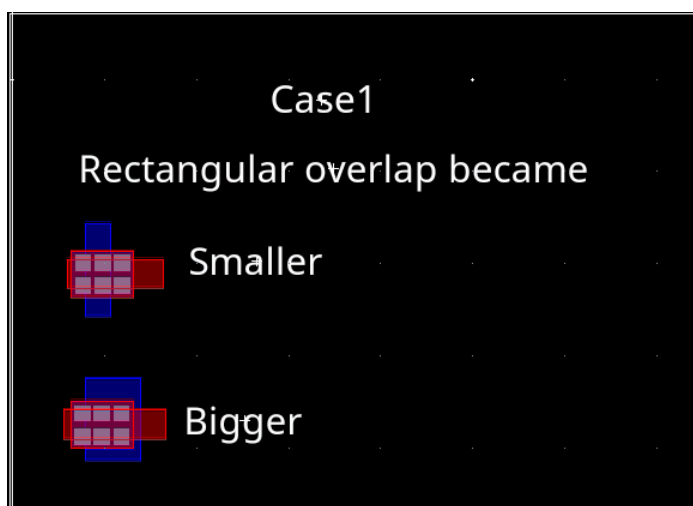
1. Open *test_lib/Fast_Interactive_Via_Editing/layout* view in EXL mode.

The Virtuoso Layout Suite EXL palette opens for editing.

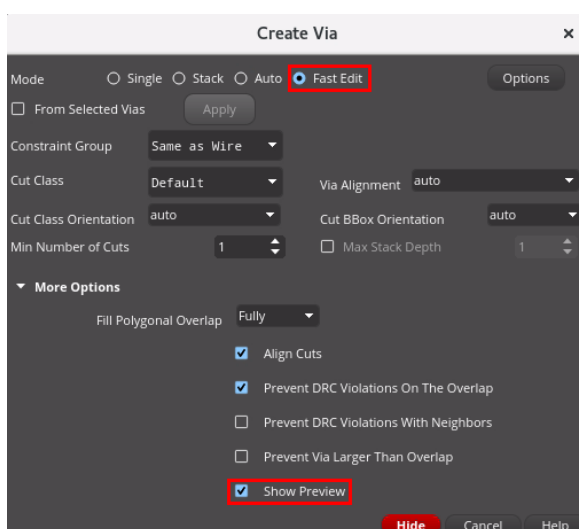


Rectangular Overlap

1. Zoom to **Case1** – Rectangular overlap became **Smaller/Bigger**.

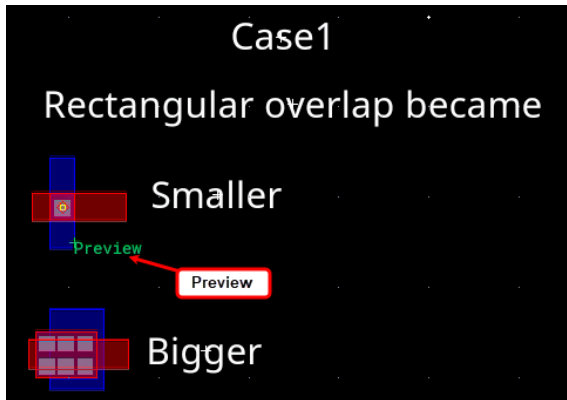


2. In the design canvas, choose the **Create – Via** menu command or use the **Create Via** icon in the Create toolbar or use the bindkey **O** to invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Fast Edit*.
 - b. Expand the **More Options Progressive Disclosure Widget**.
 - c. Select the two options: **Show Preview**.
 - d. Leave the other fields as default values.



3. In the layout canvas, move the mouse cursor over the overlap, which is smaller than the dropped via.

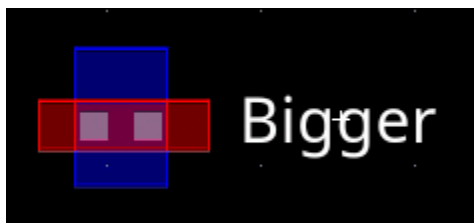
Here, you will modify the existing via that was fitting a previous overlap that became smaller. The modified via will fit the new overlap.



4. Click the **LMB** button to drop the via in the layout canvas, as shown here.



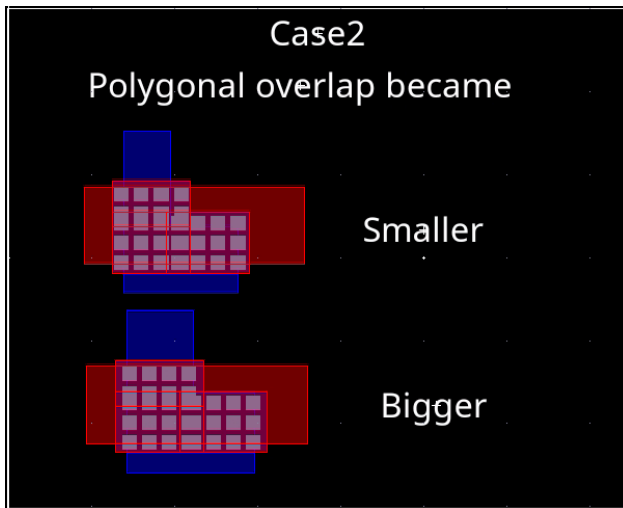
5. Repeat **Steps 4** and **5** with the overlap, which is bigger, to drop the via in the layout canvas, as shown here.



6. Press the **Esc** key to cancel the Create Via command.

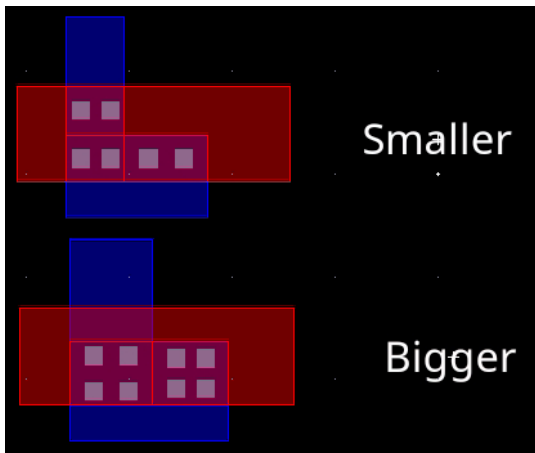
Polygonal Overlap

1. Zoom to **Case2 – Polygonal overlap became Smaller/Bigger.**



2. Invoke the **Create Via** command and set the **Mode** to *Fast Edit*.
3. Use the **Fast Interactive Via Editing (FIVE)** and edit/modify the existing vias when the overlapping area is **Smaller/Bigger**, as shown in the previous **Rectangular Overlap** section.

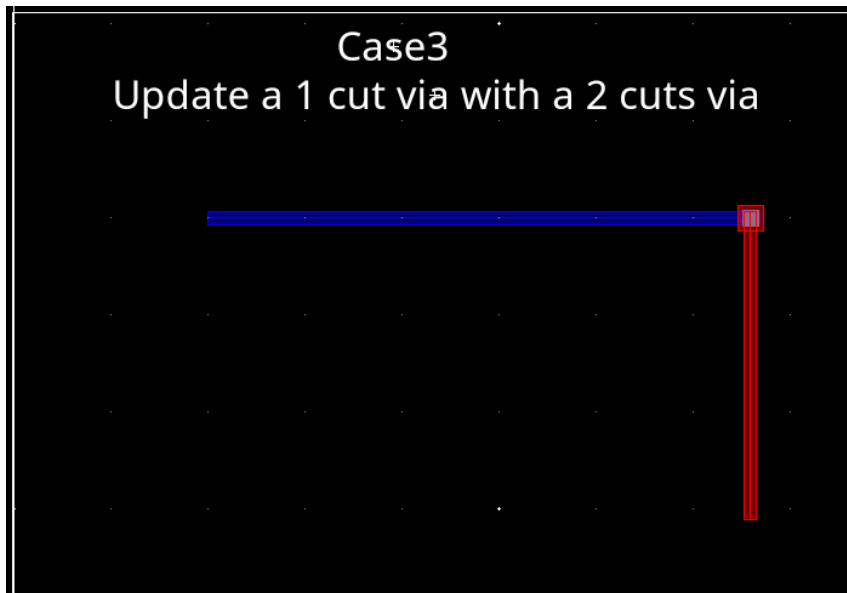
Refer to the graphic shown here.



4. Press the **Esc** key to cancel the Create Via command.

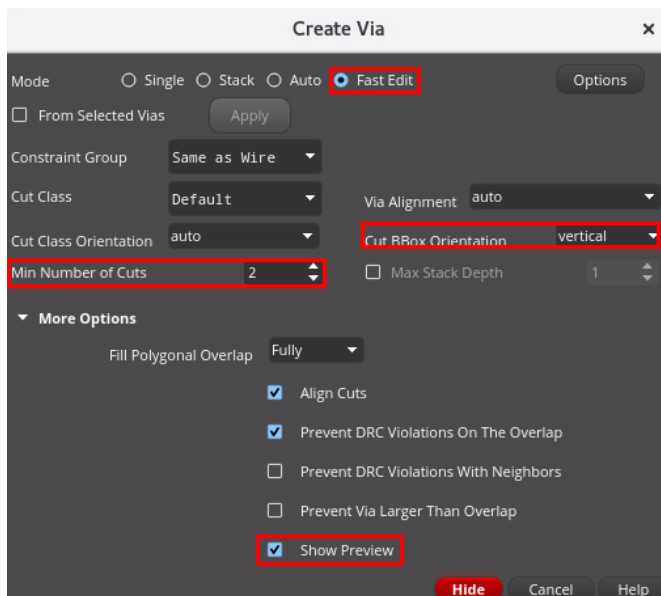
Update to Double Cut

1. Zoom to **Case3 – Update a 1 cut via with a 2 cuts via.**

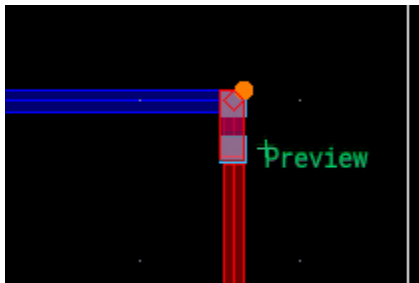


2. Invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Fast Edit*.
 - b. Set **Min Number of Cuts** = 2.
 - c. Set **Cut BBox Orientation** = **Vertical**.
 - d. Expand the **More Options** *Progressive Disclosure Widget*.
 - e. Select the option: **Show Preview**.

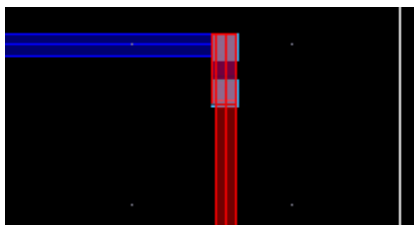
- f. Leave the other fields as default values.



3. In the design window, move the mouse cursor over the existing via to view the **Preview**.



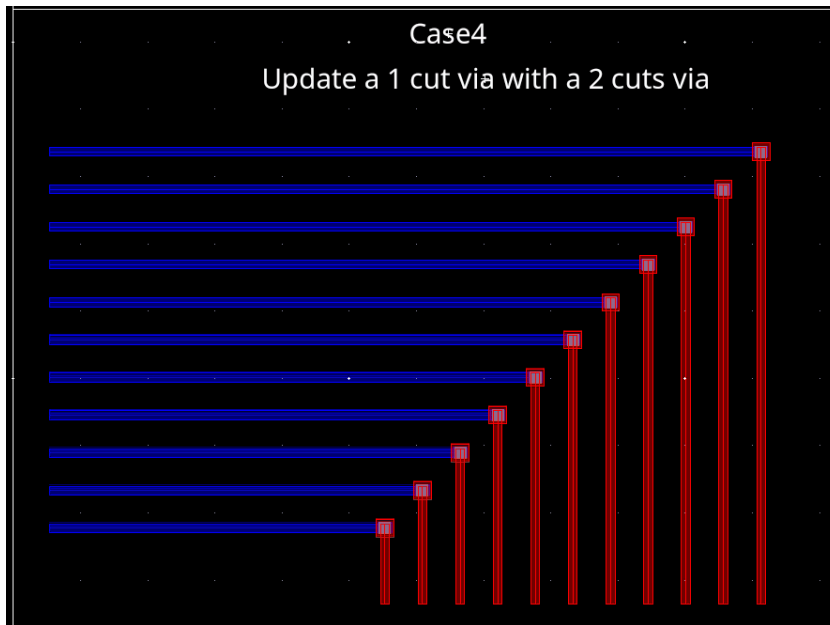
4. Click the **LMB** button to drop the via in the layout canvas, as shown here.



5. Press the **Esc** key to cancel the Create Via command.

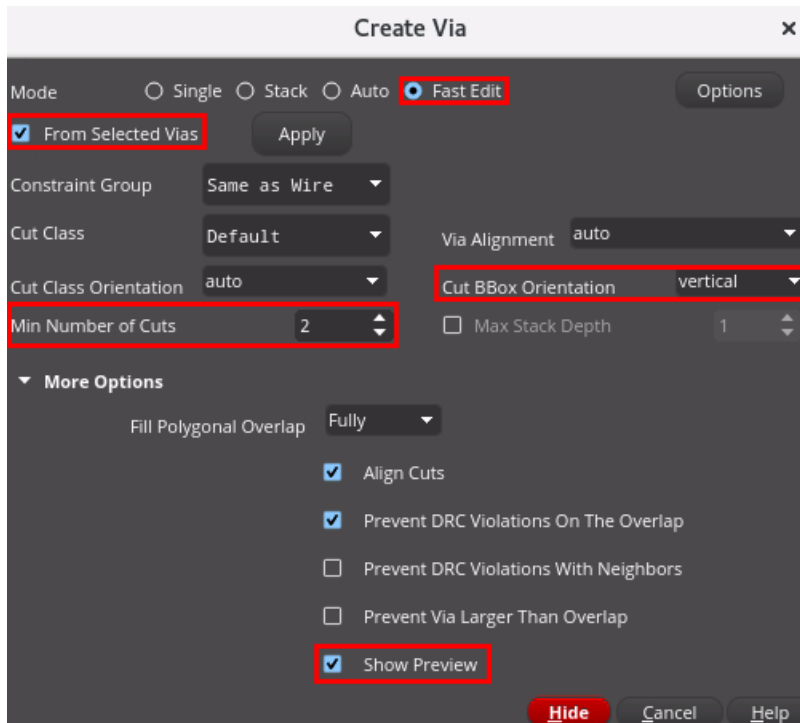
Fast Edit on Selected Vias

1. Zoom to **Case4 – Update a 1 cut via with a 2 cuts via.**

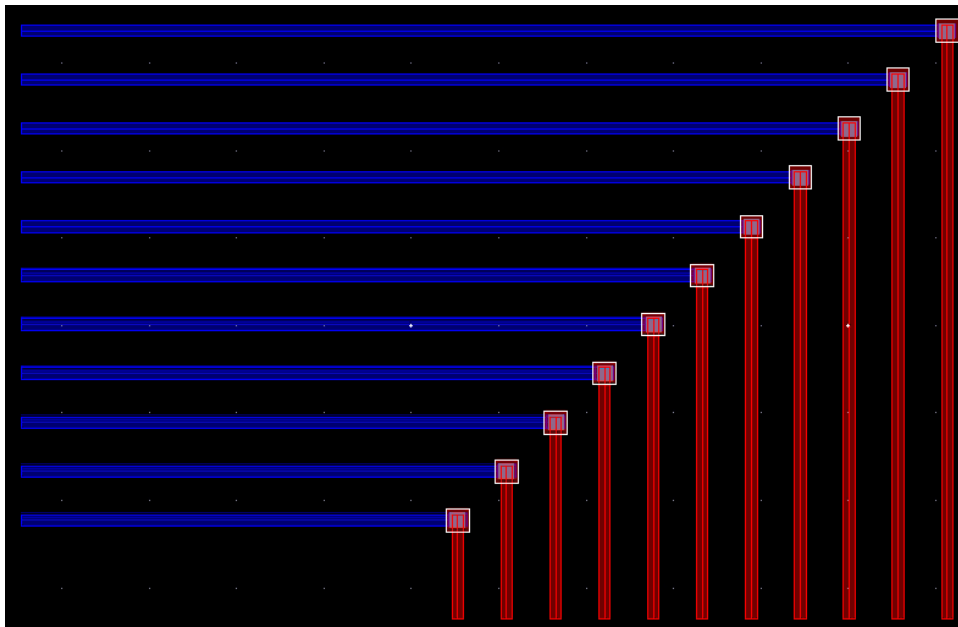


2. Invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Fast Edit*.
 - b. Enable **From Selected Vias** radio button.
 - c. Set **Min Number of Cuts** = 2.
 - d. Set **Cut BBox Orientation** = **Vertical**.
 - e. Expand the **More Options** *Progressive Disclosure Widget*.
 - f. Select the option: **Show Preview**.

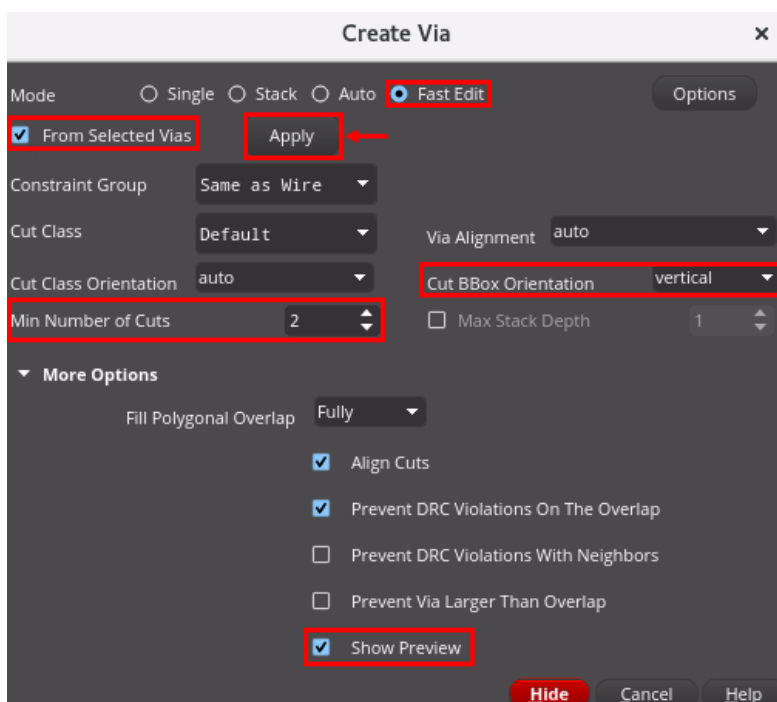
g. Leave the other fields as default values.



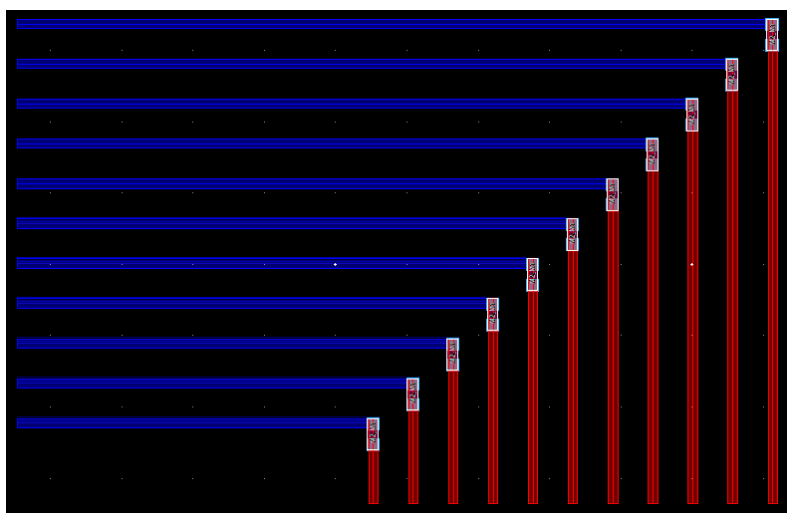
3. Area Select the vias; **11 via objects** should be selected, as shown here.



- Click the **Apply** button in the Create Via form.



The selected **11 vias** end up with **2 cuts**, as shown here.



Alternatively, you can have the same behavior by using the **MMB**. The steps are:

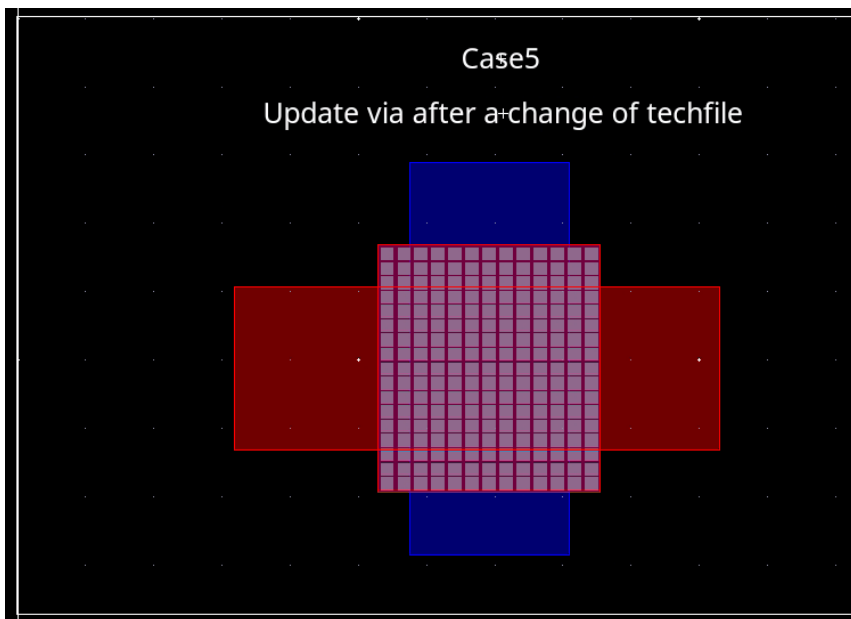
- You keep the **From Selected Vias** radio button disabled.
- You can skip it, meaning that you don't need to select the vias that you want to update.
- In the layout canvas, with the **MMB**, you draw an area surrounding the vias that you want to update. You should get the vias updated to **2 cuts**.

8. Press the **Esc** key to **Cancel** the Create Via command.

Update Via After Change of Techfile

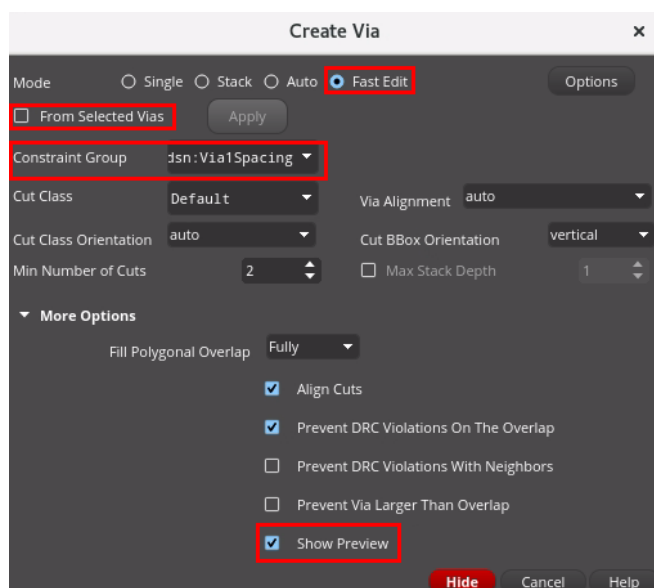
You will see an example where **Fast Edit** addresses techfile update or constraint group change.

1. Zoom to **Case5 – Update via after a change of techfile.**



2. Invoke the Create Via form and set the following values.
 - a. Set **Mode** to *Fast Edit*.
 - b. Uncheck **From Selected Vias** radio button is already checked.
 - c. Set **Constraint Group** to **dsn:Via1Spacing**.
 - d. Expand the **More Options** *Progressive Disclosure Widget*.
 - e. Select the option: **Show Preview**.

f. Leave the other fields as default values.



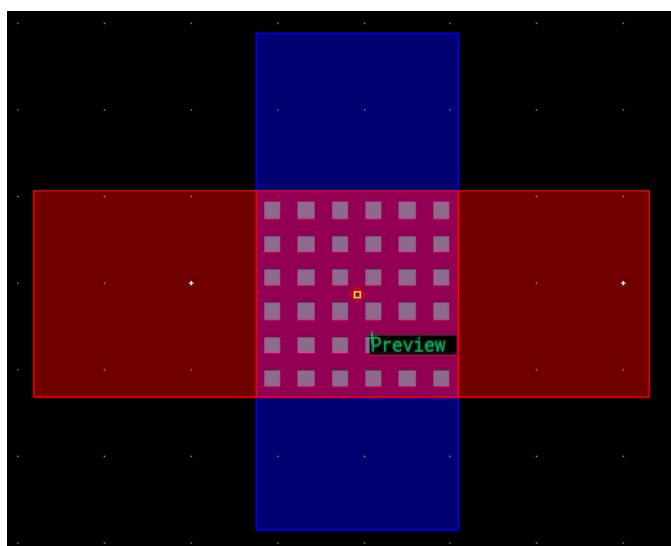
In the techfile, **via1 spacing = 0.15** whereas in the **Via1Spacing** constraint group **via1 spacing = 0.25**. The existing via was dropped with **cut spacing = 0.15**.

You can check the **Via1Spacing** constraint group in the **Process Rule Editor**.

To start the Process Rule Editor:

- On the Constraint Manager toolbar, click the **Process Rule Editor** button. 
- Alternatively, **right-click** the design object in the layout canvas or the layout navigator and choose **Process Rule Editor** from the shortcut menu.

3. In the layout canvas, move the mouse over the overlap to view the **Preview**.



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